

Understanding Robustness Limits of Learned Cardinality Estimators: A Worst-Case Data Distribution Analysis

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ABSTRACT

Learned cardinality estimators have achieved superior accuracy, yet their inherent fragility to data drift remains a primary barrier to production deployment. Existing robustness evaluations typically rely on simplified drift assumptions, failing to bound performance degradation under the complex, non-random logic of real-world workloads. This lack of reliability guarantees necessitates a worst-case analysis to determine how minor data perturbations can maximally compromise the integrity of the learned distribution.

While the susceptibility of models to data drift is widely acknowledged, identifying the minimal drift that maximizes performance degradation is fundamentally challenging: we prove that finding such a minimal drift is NP-Hard. To bridge this gap, we develop a polynomial-time algorithm with a theoretical $(1 - \kappa)$ approximation guarantee. Our analysis reveals a severe structural vulnerability: unlike prior benchmarks requiring extensive shifts (e.g., $> 20\%$) to trigger bad performance, we demonstrate that perturbing a targeted subset of merely $\approx 0.8\%$ of tuples suffices to inflate 90th-percentile Qerrors by three orders of magnitude. This shift triggers up to a $20\times$ increase in query latency on STATS-CEB and IMDB-JOB. Finally, we leverage DSA to guide five distinct robustness designs—ranging from maintenance sentinels to architectural hardening—to mitigate such worst-case degradation.

PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at <https://github.com/LiYingZe/DACA>.

1 INTRODUCTION

Cardinality Estimation (CE) plays a crucial role in query optimization within database systems, as it predicts the number of tuples that satisfy a query without execution and aids the database optimizer select the best query execution plan with the lowest cost [27], thus improving system efficiency [36, 67]. Compared to traditional estimators [52, 62], learned cardinality estimators offer more accurate estimations and have therefore garnered extensive attention in recent years [38, 48, 56, 77]. These learned approaches are generally categorized into Data-driven approaches [38, 77], Query-driven approaches [30, 48] and Hybrid approaches [56, 60, 70].

Despite their empirical success, the integration of learned estimators into production database kernels remains bottlenecked by concerns regarding their stability and robustness in dynamic environments [36, 56, 60, 81]. While existing literature suggests that learned estimators can tolerate modest distribution shifts under specific, restricted patterns (e.g., random/sequential updates) [56, 60, 67],

their behavior under complex real-world perturbation logic remains largely unexplored [68]. Consequently, the "worst-case" performance of these models under data evolution remains a black box [54, 81]. To build a truly resilient database kernel, rigorous mechanisms are required to audit and bound estimator performance against complex data drifts [26, 80].

To date, robustness research has primarily focused on workload-centric sensitivity analysis. PACE [81], proposed by Zhang et al., involves adding noise to query workloads and stress-testing the query-driven estimators. This degrades the accuracy of query-driven models and slows down the end-to-end performance. Although workload-centric analysis can significantly impair the performance of query-driven estimators, we find that they have limited impact on widely used data-driven learned estimators [38, 47, 57, 66, 71, 72, 76, 77, 88]. This is primarily because data-driven methods utilize data-level information that is independent of historical workloads to achieve robust cardinality estimation across diverse testing query distributions. Consequently, these methods can effectively withstand perturbations that target historical workloads. Therefore, workload-centric methodology is not suitable for evaluating the robustness limits of these prevalent data-driven models.

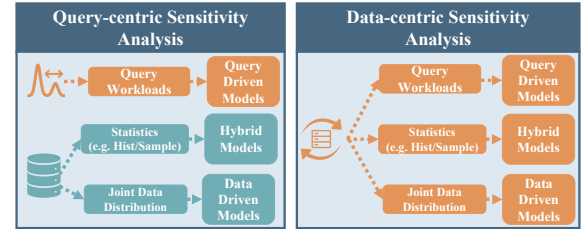


Figure 1: Query-centric vs. Data-centric Sensitivity Analysis.

Notably, data updates inevitably degrade learned cardinality estimators [50, 51, 54, 56, 66], as these models rely on static training statistics (e.g., historical workloads [30, 48] or data distributions [38, 71, 76]) that become stale as underlying properties shift [46, 51, 56]. To evaluate robustness, prior works typically rely on simplified drift simulations: Robust-MSCN [60] mimics temporal updates via sequential partitioning, while ALECE [56] employs random shuffling or broad insertions (e.g., $> 20\%$ of the entire database). However, these proxies mask the true risks in production. Real-world drifts are rarely purely random [22, 24, 45, 63] but rather complex business logic (extended discussion is in Appendix L [7]): such as flash sales [40, 58] or regional inventory purges [16, 31] which introduce biased rather than random or sequential updates. Unlike simplified assumptions, these structural shifts concentrate on specific data regions, hitting the model's 'blind spots' [54], causing catastrophic tail latencies that simplified simulations fail to expose.

This discrepancy obscures a critical blind spot: *can small-scale updates precipitate significant distribution shifts and cause performance collapses of optimizer?* Leaving this question unanswered precludes the stability guarantees essential for production adoption; users will simply reject a system prone to unpredictable latency spikes under small scale updates. Motivated by this reliability concern, we conduct a worst-case sensitivity analysis to audit how minimal but specific data drifts maximally degrade all outdated estimators.

However, constructing such a universal auditor to identify the smallest worst-case data drift is non-trivial. It requires overcoming three fundamental obstacles: **O1 The Necessity of Black-box Universality.** To uncover universal weaknesses across diverse estimators, we require an architecturally agnostic framework. Relying on white-box analyses tailored to specific mechanisms [49, 75, 81, 84] inherently limits generalization and fails in production environments where model details are opaque [8, 43]. Thus, a strict prior-free black-box assumption is essential for holistic reliability guarantees. **O2 Tractability Awareness.** Can the maximum damage of this drift be pre-calculated in polynomial-time? If computationally feasible, such analysis allows us to proactively evaluate the worst-case behavior of all learned estimators, acting as a theoretical ‘canary in a coal mine’ to alert us to potential worst-case scenarios in advance. **O3 Limited Budget.** How can a limited number of data modifications on the training data maximally degrade the accuracy of models trained on this perturbed dataset? Small-scale data modifications seem to have limited impact on learned estimators and prior studies have shown that outdated models can tolerate random data-level changes exceeding 30% while still maintaining strong estimation performance [38, 50, 67].

According to the above discussions, the toughest challenge lies in that the auditor has no information about the estimator deployed internally within the black-box, which hinders the auditor from effectively quantifying and boosting the efficiency of the stress test. To circumvent this information barrier, we leverage a pivotal empirical observation: modern learned estimators have achieved oracle-level accuracy, with over 90% of Q-errors approaching 1 [38, 56, 66, 72, 77, 88]. This phenomenon suggests that high-accuracy estimators essentially mirror the oracle. Crucially, unlike the opaque estimator, the oracle provides a distinct advantage: it possesses complete information, as its behavior is strictly determined by the underlying data distribution. This inspires us to use the oracle to mathematically “sandwich” the error range of any learned estimator. We discover that as the model learns better, this bound tightens. This leads to a counter-intuitive conclusion: high-accuracy learned estimators inherit the oracle’s sensitivities, turning their primary strength—accuracy—into a potential vulnerability.

Therefore, we can leverage this exposed common vulnerability to address the aforementioned challenge. Instead of directly tackling the black-box analysis head-on without necessary information, we can indirectly construct a finite set of data drifts designed to shift the statistics of the oracle in the training database, and thereby impact the performance of all learned estimators. Thus, to overcome O1, we utilize the oracle based on the training data as our surrogate model and convert the difficult black-box audit problem into a tuple selection problem with much more information. To tackle O2, we formulate this data-centric sensitivity analysis as a combinatorial optimization problem and establish that this problem is NP-Hard.

To achieve the most effective stress test within a limited budget and address O3, we analyze the supermodular properties of the optimization objective under constrained permutation operations and design a $(1-\kappa)$ approximation algorithm for efficient resolution.

Contributions: We make the following contributions.

C1 Black-box Sensitivity Analysis on All Estimators: We propose Data-centric Sensitivity Analysis (DSA) to identify minimal worst-case data drifts. Crucially, we formally justify this black-box transferability by proving that the oracle’s sensitivity strictly bounds the risk of any well-trained learned estimator (§ 3.2).

C2 Intractability Insights: We demonstrate that, under a limited budget, finding the optimal data drift that maximally impacts all estimators is NP-Hard, even when only considering deletion perturbation operations (§ 3.3).

C3 Near-Optimal Perturbation Strategy: We design a polynomial-time approximation algorithm (§ 4.2) that guarantees a $(1-\kappa)$ approximation for the worst-case performance of all estimators. This analysis acts as a theoretical early warning mechanism, allowing us to proactively quantify and approximate the worst-case scenarios of all learned estimators (§ 4.3).

C4 Quantifying Limits & Robustness Designs: Our experiments (§ 5) reveal a critical sensitivity gap: modifying merely 0.8% of the database triggers a 1000× error spike and raises latency by up to 20×. Beyond auditing, we utilize DSA to inspire five distinct robustness designs, transforming theoretical bounds into practical defenses against worst-case drifts.

2 PRELIMINARIES

We present learned cardinality estimators in § 2.1, stress test motivation in § 2.2, and our formal stress testing framework in § 2.3.

2.1 Learned Cardinality Estimation

Suppose a database D consists ℓ relations, i.e. $D = \{R_1, R_2, \dots, R_\ell\}$. Given a query $\mathbf{q}$, when $\mathbf{q}$ is executed on D , we obtain $\mathbf{q}$ ’s results in result set $\$(\mathbf{q}|D)$. We denote the cardinality of $\$(\mathbf{q}|D)$ as $C_D(\mathbf{q})$.

Tuple’s Joint Weight: For relation R_x with N tuples, a tuple $t_i \in R_x$ and a query $\mathbf{q}_j$, the joint weight w_{ij} denotes tuple t_i ’s contribution to query $\mathbf{q}_j$ ’s result’s cardinality, that is $w_{ij} = |\$(\mathbf{q}_j|D - R_x + \{t_i\})|$. $\mathbf{q}_j$ ’s cardinality can be seen as a linear summation of R_x ’s tuple’s the joint weights: $C_D(\mathbf{q}_j) = \sum_{i=1}^N w_{ij}$.

Learned Cardinality Estimation: Learned cardinality estimation needs to make predictions on $C_D(\mathbf{q})$ using learned estimator E without executing $\mathbf{q}$ on D . E is trained via the information of D . Based on the training information, these learned cardinality estimators can be categorized into these main paradigms: data-driven, query-driven, and hybrid.

Learned Data-driven Estimator: Learned data-driven cardinality estimator E_{Data} learns the joint distribution of database D . Based on the learned statistical information, E_{Data} provides the query independent estimation of the query $\mathbf{q}$ ’s cardinality $est(\mathbf{q}|E_{Data})$.

Learned Query-driven Estimator: Learned query-driven cardinality estimator E_{Query} learns the mapping from historical workloads Q to cardinalities C , i.e., $Q \rightarrow C$. Based on the learned mapping knowledge, E_{Query} provides the estimation of the query $\mathbf{q}$ ’s cardinality $est(\mathbf{q}|E_{Query})$.

Hybrid Estimator: Learned hybrid estimator E_{Hybrid} combines query Q with statistics (e.g. histograms [56], samples [48]) $\mathcal{S}$ as the input features, and learns the mapping $\{Q \times \mathcal{S}\} \rightarrow C$. Based on the

enhanced features, E_{Hybrid} provides a comprehensive estimation of query q 's cardinality $est(q|E_{Hybrid})$ compared to $est(q|E_{Query})$.

For query q_j , we follow the assumptions made in the majority of CE literature [36, 47, 48, 56, 76], considering q_j to be the most prevalent selection-projection-join query of the form:

```
SELECT * FROM  $q_j.Tabs$  WHERE  $q_j.Joins$  AND  $q_j.Filters$ 
```

where: $q_j.Tabs$ denotes the set of tables involved in the query q_j , such that $q_j.Tabs \subseteq D$, and $q_j.Joins$ represents the join predicates, $q_j.Filters$ signifies the filter predicates.

2.2 The Need of Active Audit: Hunting the Achilles' Heel

To ensure learned database components work reliably, robustness evaluation is essential [67], yet existing methodologies rely heavily on random or sequential drift simulations [56, 60]. While validating "average" stability, these fail to probe the performance lower bounds required for strict Service Level Agreements (SLAs) [9, 44, 83] (e.g., 99% queries < 100ms). In high-stakes sectors like finance, such worst-case "tail risks" outweigh average performance, as single violations cause severe losses [34, 64].

Crucially, real-world drifts are rarely simplified; they stem from complex business logic (e.g., flash sales [40, 58] or inventory purges [16, 31]), introducing biased updates that hit model "blind spots" [21, 54, 69] (as detailed in Appendix L [7]). For instance, updates in real-time analytics often concentrate on "hot" regions [35, 39, 55]. Ignoring these structural shifts leaves systems vulnerable to "Black Swan" events, where small scale updates trigger critical SLA violations.

However, effectively identifying these critical scenarios is challenging due to the vast space of potential data perturbations. Passive monitoring is akin to finding a needle in a haystack. We bridge this by adopting an adversarial analysis perspective [11, 23, 61], shifting from passive observers to **active auditors**. By mathematically locating the "Achilles' heel"—the specific data subset causing maximum degradation—we uncover structural vulnerabilities that random testing inherently misses.

We formally define this as Data-centric Sensitivity Analysis (DSA). Unlike PACE's [81] workload focus, DSA audits fundamental data distributions. Fundamentally, while optimizers aim to **minimize** plan costs [27], DSA seeks to **maximize** plan cost induced by estimation errors. This reveals a dangerous asymmetry: minute updates can trick optimizers into abandoning cheap operations (e.g., Hash Joins) for expensive ones (e.g., Nested Loops), breaking the system's SLA. The practical necessity of detecting such critical sensitivities is demonstrated below:

EXAMPLE 2.1. Figure 2 illustrates the core mechanism of DSA. Consider estimators initially trained on a stale database snapshot (Step 1). To probe the fundamental sensitivity limit, instead of random updates, we identify a single, targeted tuple update (setting $R_x.ID = 2$) in Step 2. Crucially, while a random update might be statistically negligible, this targeted update hits the model's specific knowledge boundary. As a result, in Step 3, the estimators remain anchored to outdated statistics (predicting a cardinality near 0), failing to adapt to the critical data change. This discrepancy misleads the query optimizer in Step 4, causing it to erroneously favor a Nested Loop join over a more efficient Hash Join. The consequence, shown for JOB-Q60 in Figure 2 (right), is severe: the algorithmic complexity degrades from the Hash Join's optimal $O(M + N)$ to the Nested Loop's prohibitive $O(M \times N)$, exploding the execution time by

over 9.2×10^4 seconds. This case demonstrates that system fragility is not defined by how much data changes, but by which data changes. DSA allows us to identify these critical weak points.

2.3 Stress Testing Framework

In this section, we establish the stress testing framework via worst-case sensitivity analysis. We formalize the model by defining the auditor, their goals, knowledge, and sensitivity evaluation metrics. **Auditor and Auditor's Goal:** To investigate the worst-case performance of all cardinality estimators in dynamically changing databases, we postulate a virtual auditor performing a stress test that emerges when the estimator's information becomes outdated. To uniformly analyze the impact of different perturbation strategies under consistent distributions, we fix the updated database state as D . The auditor aims to find minimized critical modifications ΔD (insertions/deletions) such that models trained on the perturbed distribution $D' = D + \Delta D$ yield catastrophically inaccurate estimations when applied to the original version D . This misleads the optimizer to select physical plans with the highest cost when evaluated on the original version D . For brevity, we refer to D as the *original state* and D' as the *perturbed state*. We follow the established convention of query optimization research [25, 32, 33] by interpreting each base relation under standard SQL bag semantics. In this context, tuples may appear with specific multiplicities, and a perturbation ΔD is permitted to consist of the insertion or deletion of logical duplicates. **Auditor's Knowledge:** In this work, we study a worst-case sensitivity analysis setting where the auditor (e.g., a system administrator or a robustness testing tool) has read-only access to the original database snapshot D and the schema. The auditor has no information about the internal architecture or parameters of the deployed learned estimator, except that it is trained on a perturbed snapshot D' . The auditor is given a test workload W_{Test} (defined below) and can execute diagnostic aggregate queries on D to compute the statistics needed for sensitivity analysis.

Auditor's Capacity: We assume the auditor can issue diagnostic SQL queries on D . For each query $q_j \in W_{Test}$ and a target relation R_x with primary key PK_x , the auditor can obtain joint weight w_{ij} of each tuple $t_i \in R_x$ by rewriting q_j into a counting query below:

```
SELECT  $R_x.PK_x$ , COUNT(*) FROM ... GROUP BY  $R_x.PK_x$ ,
```

which returns, for each t_i , the number of output tuples contributed by t_i to q_j on D . This capability does not require tuple lineage from the final query result; it only assumes standard read-only privileges to run aggregate diagnostics, which is realistic for an auditor. Additionally, the auditor can apply at most K update operations (insertions/deletions) to a specific relation R_x to form a perturbed snapshot $D' = D + \Delta D$ for evaluating robustness under a limited budget. When the production database enforces primary-key or foreign-key constraints, we assume the auditor operates on an offline copy or shadow tables where such multiplicities are encoded via surrogate identifiers or relaxed integrity constraints; this ensures that query answers and optimizer behavior are preserved while our stress test manipulates tuple multiplicities. We explicitly restrict modifications to a single relation to isolate the causal impact of data drift from the confounding factors of complex multi-table join interactions. While multi-table perturbations represent a broader challenge, demonstrating that catastrophic regression can

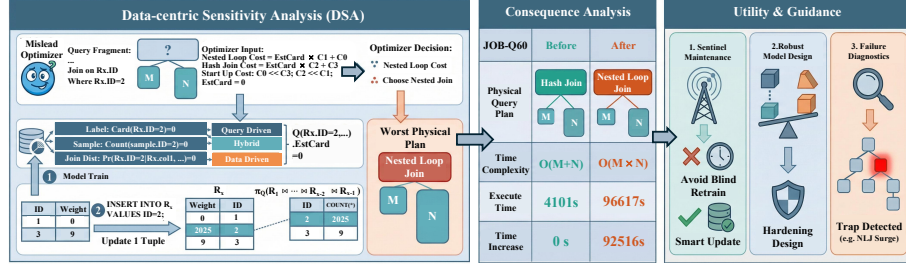


Figure 2: The Data-centric Sensitivity Analysis (DSA) Ecosystem. Illustrating the process of auditing worst-case data drifts (Left), quantifying their impact on query execution (Middle), and leveraging these insights for robust system guidance (Right).

be induced even under such limited conditions is sufficient to expose the fundamental fragility of the estimators. This design choice provides a rigorous, interpretable lens for precise root-cause analysis. **Sensitivity Evaluation Metric.** The auditor has to devise specific strategies to impair these commonly used metrics: (1) **Qerror metric:** Defined as $Q(est(q|E), C_D(q)) = \max(\frac{est(q|E)}{C_D(q)}, \frac{C_D(q)}{est(q|E)})$, where $est(q|E)$ is the estimators E 's prediction on given query q . It measures the distance between the estimated cardinality $est(q|E)$ and the true cardinality $C_D(q)$ of a query. (2) **End-to-end latency:** Measures the total latency for generating a plan with estimated cardinalities and executing the physical plan. This metric is essential for demonstrating how an estimator can improve query optimization performance. It serves as a gold standard for assessing the effectiveness of CE approaches [36, 47, 56, 71].

3 BLACK-BOX SENSITIVITY ANALYSIS FOR LEARNED ESTIMATORS

In this section, we will apply the concepts of data-centric sensitivity analysis and the stress testing framework established in § 2 to define the black-box data-centric sensitivity analysis in § 3.1. We will appropriately transform this problem into a constrained integer nonlinear programming problem in § 3.2 and analyze its computational intractability in § 3.3.

3.1 Black-box Data-centric Sensitivity Analysis

In this section, we give the original black-box Data-centric Sensitivity Analysis (DSA) problem and introduce the challenge associated with directly solving this problem. We assume that the auditor can execute at most K tuple-level data modifications ΔD on the original database state D , simulating a data drift scenario where the training state shifts to $D' = D + \Delta D$. The auditor aims to ensure that, when evaluated on the testing workload $W_{Test} = \{q_j | 1 \leq j \leq M\}$ with M queries, the estimator E , trained on the perturbed database D' , has the worst accuracy on the original database D , i.e., maximize Eq. 1:

$$\text{Max} : \sum_{j=1}^M Q(est(q_j|E), C_D(q_j)) = \sum_{j=1}^M \max \left(\frac{est(q_j|E)}{C_D(q_j)}, \frac{C_D(q_j)}{est(q_j|E)} \right) \quad (1)$$

$$\text{s.t. (1)} D' = D + \Delta D \quad (2) E \text{ is trained on } D' \quad (3) |\Delta D| \leq K$$

When maximizing Eq. 1, we assume a universal scenario, where the auditor don't need to acquire knowledge of the internal estimator type used in the database. Specifically, while the auditor knows that E is trained on D' , they are unaware of whether E is data-driven, query-driven, or hybrid. Additionally, the black-box setting restricts the auditor's visibility by concealing details of the deployed model, introducing significant challenges. The internal

models could include MLPs [30, 77], CNNs [48, 60], Transformers [56, 76], or Bayesian networks [71, 72]. Consequently, the auditor cannot leverage existing white-box evaluation methodologies from ML security [13, 17, 18, 42, 73, 74, 82]. This renders direct black-box data-centric DSA seemingly infeasible, necessitating a novel approach to reframe the problem.

3.2 Targeting the Surrogate Oracle

To solve the model-agnostic in the previous section, we observe that modern estimators increasingly converge toward oracle-level accuracy [36, 56, 67, 76, 77]. Meanwhile, under the premise of model-agnosticism, the oracle provides a richer informational surrogate, offering a direct solution to the model-agnostic challenge identified in the previous chapter. We therefore focus our robustness analysis on the oracle estimator (E_O) as the representative surrogate.

The Oracle-to-Learned Connection. A critical question arises: does a worst-case sensitivity in the theoretical oracle E_O translate to a concrete learned estimator E ? To answer this, we formally derive a transfer bound. Driven by the fact that cardinality is naturally bounded by the dataset size, and consistent with empirical evidence reporting finite Max Qerrors in prior works [48, 76, 88], we assume the learned estimator E has a bounded factor α (i.e., $\forall q \in W, Q(est(q|E), est(q|E_O)) \leq \alpha$). We now present the following to bound the aggregate error of the learned model using the oracle's error. (The proof of Th 3.1 and Th 3.2 is in Appendix K [7])

THEOREM 3.1 (SANDWICHED BOUND). *Let E be a learned estimator trained on D' with a factor α . The aggregate Qerror of E on the original database D is sandwiched by the oracle's aggregate error:*

$$\frac{1}{\alpha} \sum_{q \in W} Q(est(q|E_O), C_D(q)) \leq \sum_{q \in W} Q(est(q|E), C_D(q)) \leq \alpha \sum_{q \in W} Q(est(q|E_O), C_D(q))$$

Theorem 3.1 confirms that the oracle's aggregate Qerror mathematically bounds the estimator's error.

The Cost of Learning. One might question whether α is small enough in practice to maintain this tight link. We provide a theoretical guarantee covering the broad class of models trained via log-space regression [30, 48, 56, 60].

THEOREM 3.2 (LOG-SPACE ALIGNMENT). *If model E minimizes the standard log-space loss (i.e., $|\log est(q|E) - \log est(q|E_O)| \leq \eta$), then its bound factor is strictly bounded by $\alpha \leq e^\eta$. This implies that minimizing log-space error is equivalent to minimizing α .*

Theorem 3.2 reveals that: the "better" a model learns (i.e., lower log-space error $\eta \rightarrow 0$), the closer α approaches 1, and the more inescapably it inherits the oracle's sensitivity to data drifts.

Consequently, our objective simplifies to maximizing the Qerror of the oracle on D' , which serves as a worst-case certificate for all high-capacity learned estimators:

$$\text{Max} : \sum_{j=1}^M \max \left(\frac{\text{est}(\mathbf{q}_j|E_O)}{C_D(\mathbf{q}_j)}, \frac{C_D(\mathbf{q}_j)}{\text{est}(\mathbf{q}_j|E_O)} \right) = \sum_{j=1}^M \max \left(\frac{C_{D'}(\mathbf{q}_j)}{C_D(\mathbf{q}_j)}, \frac{C_D(\mathbf{q}_j)}{C_{D'}(\mathbf{q}_j)} \right) \quad (2)$$

$$\text{s.t. (1)} D' = D + \Delta D \quad (2) |\Delta D| \leq K$$

Given the test workload W_{Test} and a relation R_x of N tuples, denoted as $R_x = \{t_1, t_2, \dots, t_N\}$. We consider using $t_i.\beta$ to represent the auditor's behavior on tuple t_i , where $t_i.\beta$ takes -1 to indicate delete tuple t_i from the R_x , $t_i.\beta$ takes 0 to indicate no operation on tuple t_i , and $t_i.\beta$ takes k to indicate duplicating tuple t_i by k times and re-inserting them into the database. Therefore, for the database D' , the cardinality of query $\mathbf{q}_j$ on D' equals to $C_{D'}(\mathbf{q}_j) = C_D(\mathbf{q}_j) + \sum_{i=1}^N t_i.\beta \times w_{ij}$. Based on the above, we reorganize the sensitivity analysis problem in Eq. 1 into:

Optimal Data-centric Sensitivity Analysis (Optimal DSA)

Problem: Given testing workloads W_{Test} , the auditor needs to provide a modification strategy and maximize the following Eq. 3:

$$\text{Max} : \sum_{j=1}^M \max \left(\frac{C_D(\mathbf{q}_j) + 1 + \sum_{i=1}^N t_i.\beta \times w_{ij}}{C_D(\mathbf{q}_j) + 1}, \frac{C_D(\mathbf{q}_j) + 1}{C_D(\mathbf{q}_j) + 1 + \sum_{i=1}^N t_i.\beta \times w_{ij}} \right) \quad (3)$$

$$\text{s.t. } \sum_{i=1}^N |t_i.\beta| \leq K \quad t_i.\beta \in \{-1, 0, 1, \dots, K\}$$

For the sake of brevity, we employ the abbreviation 'optimal DSA' to represent 'optimal Data-centric Sensitivity Analysis' throughout the remainder of this paper. Meanwhile, we add 1 to both the numerator and denominator when calculating Qerror to avoid division by zero errors. This transformation is a commonly seen evaluation technique in many cardinality estimation methods and can be found in many open-sourced cardinality estimators' GitHub repositories e.g. line 4 in [5], line 21-22 in [6].

In summary, we converted the seemingly unsolvable black-box sensitivity analysis problem stated in Eq. 1 into a constrained integer nonlinear programming problem, as detailed in Eq. 3. However, it remains uncertain whether the auditor can effectively solve Eq. 3 in polynomial-time. Therefore, we will provide an analysis of intractability in the next section.

3.3 Intractability

In this section, we give an intractability analysis of the optimal DSA problem defined in Eq. 3. We conclude that finding the optimal DSA solution is NP-Hard even when considering deletions alone. Additionally, when considering both insertions and deletions, finding an optimal strategy still remains NP-Hard. When only considering data deletions, the problem is converted into the following optimization problem in Eq. 4:

$$\begin{aligned} \text{Max: } & \sum_{j=1}^M \frac{C_D(\mathbf{q}_j) + 1}{C_D(\mathbf{q}_j) + 1 + \sum_{i=1}^N t_i.\beta \times w_{ij}} \\ \text{s.t. } & \sum_{i=1}^N |t_i.\beta| \leq K \quad t_i.\beta \in \{-1, 0\} \end{aligned} \quad (4)$$

We now present a proof sketch that finding the optimal solution to Eq. 4, is NP-Hard. We start our polynomial-time reduction from the known Densest- K -Subgraph problem, defined as:

Densest- K -Subgraph (DKS) problem: Given a simple, undirected graph $\mathbb{G} = (\mathbb{V}, \mathbb{E})$ and an integer K , find a subset of vertices $S \subseteq \mathbb{V}$ such that: $|S| \leq K$ and the subgraph $\mathbb{G}[S]$ induced by S has the maximum number of edges.

The left subfigure in Figure 3 shows a DKS example when $K = 4$. The DKS problem is a well-studied NP-Hard problem in graph theory [10, 19, 20] and closely related to many applications in graph database community such as community search [85–87]. We will next construct a polynomial-time reduction based on this problem.

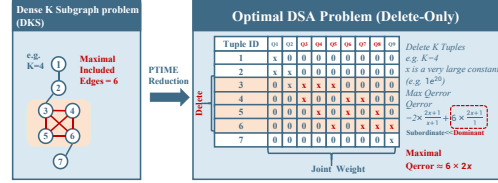


Figure 3: Polynomial-time reduction example starting from Densest- K -Subgraph(DKS) problem.

THEOREM 3.3. *The optimal DSA problem defined in Eq. 4 when considering only deletions, is NP-Hard.*

PROOF. (Sketch) The polynomial-time reduction is as follows: Given a DKS instance $\mathbb{G} = (\mathbb{V}, \mathbb{E})$ and integer K , we construct a DSA instance in $O(|\mathbb{V}| \cdot |\mathbb{E}|)$ time: (1) Each vertex v_i maps to a tuple $t_i \in R_x$. (2) Each edge $e_j = \{v_{i1}, v_{i2}\}$ maps to a query $\mathbf{q}_j$ covering exactly tuples t_{i1} and t_{i2} . (3) We assign each tuple a common join weight x , where x is a sufficiently large positive integer.

For the constructed delete-only DSA problem P_2 , and a given query $\mathbf{q}_j$ consisting of two tuples. If one of its tuples is deleted, the contribution to the objective function is $\Delta Q_{err1} = \frac{2x+1}{x+1}$, if two tuples are both deleted, the contribution is $\Delta Q_{err2} = 2x + 1$. When x is sufficiently large, ΔQ_{err1} approaches 2 and ΔQ_{err2} approaches $2x$ which is significantly larger: ($\Delta Q_{err2} \gg \Delta Q_{err1}$). Consequently, the maximization objective is predominantly influenced by ΔQ_{err2} . This implies that maximizing the objective function is effectively equivalent to maximizing $2x$ times the number of edges in the K -densest subgraph of $\mathbb{G}$. To better illustrate the above ideas, we use Example 3.1 to show the reduction process.

EXAMPLE 3.1. Figure 3 presents a reduction example. The left side depicts a DKS problem instance with $K = 4$, while the right side shows a corresponding optimal delete-only DSA problem for $K = 4$. For a graph with 7 vertices and 9 edges, we construct a database instance with a relation R_x containing 7 tuples and 9 queries. For instance, queries $\mathbf{q}_3$, $\mathbf{q}_4$, and $\mathbf{q}_5$ encode connections between vertex 3 and vertices {4, 5, 6}. Each tuple contributes x to its associated query's cardinality. When x is sufficiently large (e.g., 10^{20}), deleting one tuple for a single query yields a Qerror contribution of approximately 2, while deleting two tuples results in $2x$. Thus, the maximum Qerror arises when tuples from two queries are deleted. Moreover, the worst-case Qerror contribution scales with the number of internal edges in the K -vertex subgraph selected by DKS. In our example, the optimal DKS solution is a 4-vertex clique with 6 edges, leading to a maximum Qerror of $6 \times 2x$ for the delete-only DSA.

Since the DKS problem is NP-Hard, and we have reduced it to the optimal delete-only DSA problem in polynomial-time, it follows that finding the optimal DSA problem when only considering deletions is NP-Hard. $\square$

Based on the Theorem 3.3, we show that considering full insertion and deletion, the complete DSA problem is also NP-Hard.

THEOREM 3.4. *The optimal DSA problem defined in Eq. 3 when considering both insertions and deletions, is NP-Hard.*

PROOF. (Sketch): Our overall proof sketch is as follows: We extend the polynomial-time reduction constructed in Theorem 3.3. When the joint weight $\mathbf{x}$ is larger than $K \times M$, where K is the perturbation budget, and M is the number of queries, the benefits from I insertion operations ($2 \leq I \leq K$) become negligible compared to deleting two tuples within a given query $\mathbf{q}_j$. This leads the auditor to abandon insertion operations and transform the case into a delete-only environment, and we can use the remainder of Theorem 3.3's proof. Therefore, we establish a polynomial-time reduction from the DKS problem to the optimal DSA problem that considers both insertions and deletions, thereby proving that the optimal perturbation strategy involving both insertions and deletions is NP-Hard. More rigorous proof is in Appendix A [7]. $\square$

In summary, computing the exact optimal solution to the DSA problem is NP-Hard, meaning that a scalable auditor cannot, in general, certify the worst-case drift by exhaustive search in polynomial time. This result characterizes an inherent computational barrier for worst-case sensitivity analysis and motivates the need for principled approximations. This result motivates us to focus on efficient approximation algorithms: as shown next, auditors can design polynomial-time algorithms with provable guarantees to obtain near-optimal worst-case scenarios.

4 NEAR-OPTIMAL SENSITIVITY ANALYSIS

While the previous section showed that finding universally worst-case perturbations is NP-Hard, the DSA problem possesses structural properties that enable efficient solutions. Specifically, § 4.1 analyzes these properties, which we then leverage in § 4.2 to design an approximate algorithm. Finally, § 4.3 provides a rigorous performance analysis of the proposed method.

4.1 Analysis of the DSA's Objective Function

In this section, we will analyze the nature of DSA's objective function under specific conditions to derive certain insights. To facilitate the presentation of symbols in the following discussion, we denote the Qerror induced by the j -th query, considering only deletions, as: $\mathbb{Q}_j(X) = \frac{C_D(\mathbf{q}_j)+1}{C_D(\mathbf{q}_j)+1+\sum_{t_i \in X} t_i \cdot \beta \times w_{ij}}$. X is the deleted tuples within the relation R_X , having $X \subseteq R_X$, $\forall t_i \in X$, $t_i \cdot \beta = -1$. Therefore, the total Qerror when only deletion is considered is $\mathbb{Q}(X) = \sum_{j=1}^M \mathbb{Q}_j(X)$. Based on these definitions, we have Theorem 4.1

THEOREM 4.1. *When only deletion operations are considered, total Qerror $\mathbb{Q}(X)$ of the optimal DSA problem satisfies the supermodular property, that is: $\forall A, B \subseteq R_X$, $\mathbb{Q}(A \cup B) + \mathbb{Q}(A \cap B) \geq \mathbb{Q}(A) + \mathbb{Q}(B)$.*

PROOF. (Sketch) We aim to demonstrate that the objective

$$\mathbb{Q}(X) = \sum_{j=1}^M \mathbb{Q}_j(X) = \sum_{j=1}^M \frac{C_D(\mathbf{q}_j) + 1}{C_D(\mathbf{q}_j) + 1 + \sum_{t_i \in X} t_i \cdot \beta \times w_{ij}}$$

satisfies the supermodular property. Specifically, for any two sets A and B within R_X , we need to prove that $\forall A, B \subseteq R_X$, $\mathbb{Q}(A \cup B) + \mathbb{Q}(A \cap B) \geq \mathbb{Q}(A) + \mathbb{Q}(B)$.

First, consider each component function $\mathbb{Q}_j(X)$. We aim to show that $\mathbb{Q}_j(X)$ is supermodular. We denote:

$$w'_{ij} = \frac{w_{ij}}{1 + C_D(\mathbf{q}_j)}, \quad S_j(X) = \sum_{t_i \in X} t_i \cdot \beta \times w'_{ij}$$

Then, the expression:

$$\mathbb{Q}_j(A \cup B) + \mathbb{Q}_j(A \cap B) - \mathbb{Q}_j(A) - \mathbb{Q}_j(B)$$

can be reorganized into the following form, we provide the detailed derivation in the Appendix B [7].

$$\frac{(2 + S_j(A) + S_j(B))(S_j(A \setminus B))(S_j(B \setminus A))}{(1 + S_j(A \cup B))(1 + S_j(A \cap B))(1 + S_j(A))(1 + S_j(B))}$$

Observe that given $\forall t_i \in X$, $t_i \cdot \beta = -1$, $w'_{ij} \geq 0$, we have $S_j(X)$ is negative. We deduced that the two factors of the numerator, $S_j(A \setminus B)$ and $S_j(B \setminus A)$, are both less than or equal to 0. Thus, their product is greater than or equal to 0. Meanwhile, $\forall X \subseteq R_X$, $(1 + S_j(X)) \geq (1 + S_j(R_X)) = \frac{1}{1 + C_D(\mathbf{q}_j)} > 0$. Therefore, each term in the denominator is strictly positive, and $(2 + S_j(A) + S_j(B))$ is strictly positive. In conclusion, within the above fractional, the numerator is greater than or equal to 0, while the denominator is strictly greater than 0. We have:

$$\mathbb{Q}_j(A \cup B) + \mathbb{Q}_j(A \cap B) - \mathbb{Q}_j(A) - \mathbb{Q}_j(B) \geq 0,$$

which confirms that $\mathbb{Q}_j(X)$ is supermodular. Since $\mathbb{Q}(X)$ is the sum of supermodular functions $\mathbb{Q}_j(X)$, it inherits the supermodular property. Formally,

$$\begin{aligned} & \mathbb{Q}(A \cup B) + \mathbb{Q}(A \cap B) - \mathbb{Q}(A) - \mathbb{Q}(B) \\ &= \sum_{j=1}^M [\mathbb{Q}_j(A \cup B) + \mathbb{Q}_j(A \cap B) - \mathbb{Q}_j(A) - \mathbb{Q}_j(B)] \geq 0. \end{aligned}$$

Hence, the objective $\mathbb{Q}(X)$ satisfies the supermodular property. $\square$

Next, we consider the scenario where only insertions are considered, the DSA problem discussed in Eq. 3 can be much simpler. Similar to Theorem 4.1, we define the Qerror induced by the j -th query, considering only insertions, as: $\mathbb{Q}'_j(X) = \frac{C_D(\mathbf{q}_j)+1+\sum_{t_i \in X} t_i \cdot \beta \times w_{ij}}{C_D(\mathbf{q}_j)+1}$. And X is the set of tuples that are repeatedly inserted into R_X , having $X \subseteq R_X$, $\forall t_i \in X$, $t_i \cdot \beta = 1$. Therefore, the total Qerror when only insertion is considered is $\mathbb{Q}'(X) = \sum_{j=1}^M \mathbb{Q}'_j(X)$. Based on these definitions, we have Theorem 4.2:

THEOREM 4.2. *When only considering insertions, the optimal DSA problem satisfies the modular property, i.e., $\forall A, B \subseteq R_X$, $\mathbb{Q}'(A \cup B) + \mathbb{Q}'(A \cap B) = \mathbb{Q}'(A) + \mathbb{Q}'(B)$.*

PROOF. We leave the detailed proof in our Appendix C [7]. $\square$

The above theorems indicate that although the optimal DSA problem in Eq. 3 is NP-Hard, the objective function possesses special properties when the modification types are limited. We will utilize this characteristic to design an effective approximate perturbation algorithm and analyze the stress test's effectiveness.

Algorithm 1 Approximate solution to optimal DSA.

Require: Table R_x ; Testing queries W_{Test} ; Results $\mathcal{R}_x$; Budget K ;
Ensure: Modification set ΔD with at most K modifications;

```

1:  $w = \text{getJointWeight}(\mathcal{R}_x, R_x)$  ▷ Obtaining joint weight
2:  $\Delta D = \phi; \text{Mask} = 0_N$ 
3: while  $|\Delta D| \leq K$  do
4:    $\text{bestOp} = \phi; \text{bestVal} = 0;$ 
5:   for  $(t_i \in R_x) \wedge (\text{Mask}[i] = 0)$  do ▷ Get  $R_x$ 's remained tuples
6:      $\text{gainD} = \sum_{q_j \in W_{Test}} Q(C_{D+\Delta D}(q_j), C_{D+\Delta D}(q_j) - w_{ij});$ 
7:     if  $\text{gainD} \geq \text{bestVal}$  then
8:        $\text{bestVal} = \text{gainD}$ 
9:        $\text{bestOp} = -t_i$ 
10:     $\text{gainI} = \sum_{q_j \in W_{Test}} Q(C_{D+\Delta D}(q_j), C_{D+\Delta D}(q_j) + w_{ij});$ 
11:    if  $\text{gainI} \geq \text{bestVal}$  then
12:       $\text{bestVal} = \text{gainI}$ 
13:       $\text{bestOp} = t_i$ 
14:    if  $\text{bestOp} = \phi$  then
15:      break
16:    if  $\text{bestOp} = -t_\delta$  then ▷ Mask the deleted tuple
17:       $\text{Mask}[\delta] = 1$ 
18:     $\Delta D.append(\text{bestOp});$ 
19: return  $\Delta D;$ 

```

4.2 Perturbation Strategy Generation

In this section, we will utilize the characteristics analyzed in the previous section to design a corresponding algorithm to maximize Eq. 3. We first present the perturbation strategy in Algorithm 1:

First, the auditor computes the joint weight of each tuple in R_x on the query results $\mathcal{S}(q_j|D)$ (Line 1) via the following SQL:

SELECT $R_x.PK, \text{COUNT}(\ast)$ FROM $\mathcal{S}(q_j|D)$ GROUP BY $R_x.PK$;

where $R_x.PK$ is the primary key of the relation R_x . Lines 2 and 4 of the algorithm handle the initialization of variables. Lines 5-13 iterate through the modified relation table one by one, computing the weights for both deletion and duplicate insertion for each tuple. In each iteration, the algorithm greedily selects the operation that maximizes the local Qerror. Lines 16-17 temporarily save the deleted tuples to prevent duplicate computation on already deleted tuples. The algorithm guarantees termination with a time complexity $O(|\mathcal{R}_x| + N \times K \times M)$, where: $|\mathcal{R}_x|$ is the overhead of the group by operation in line 1 and is linear to the size of the result set $\mathcal{R}_x$, N is the cardinality of the relation R_x , K is the budget allocated to the auditor for data insertion, and M is the size of the test query.

Crucially, beyond constructing the worst-case limit, Algorithm 1 serves a practical diagnostic role. Due to the monotonic and super-modular nature of the objective function, even a partial overlap between real-world updates and the identified set ΔD induces significant accuracy degradation. This property implies that the identified high-sensitivity tuples can be monitored as ‘‘Sentinels’’ for early warning, a strategy we theoretically justify in Appendix I&L and experimentally validate in Section 5.5.

4.3 Analysis on the Perturbation Strategy

This section evaluates the perturbation strategy in Algorithm 1. We show it achieves optimal or near-optimal results when operations are limited to either insertions or deletions. Furthermore, we provide two corollaries for general scenarios and prove that the algorithm maintains a $1 - \kappa$ approximation ratio in deletion-only cases.

THEOREM 4.3. *In the scenario where only deletions are considered, let $Q(O)$ denote the optimal DSA result. Algorithm 1 can achieve an*

approximation of $(1 - \kappa)$. That is, the output ΔD of the Algorithm 1 possesses the property that $Q(\Delta D) \geq (1 - \kappa)Q(O)$, where $\kappa = 1 - \min_{t \in R_x} \min_{A, B \subseteq R_x - t} \frac{Q(A+t) - Q(A)}{Q(B+t) - Q(B)}$.

PROOF. (Sketch). Based on the definition of κ , we obtain $\forall A, B \subseteq R_x, 1 - \kappa \leq \min_{t \in R_x} \frac{Q(A+t) - Q(A)}{Q(B+t) - Q(B)}$. Furthermore, we let the set ΔD_i denote the set chosen by Algorithm 1 at the i -th step, and we let ΔD_i to substitute A . Similarly, we can arbitrarily choose a subset permutation of the optimal solution O and let a subset O_i with size i from the permutation to substitute B . We also define $t_{i1} = O_{i+1} - O_i$ and $t_{i2} = \Delta D_{i+1} - \Delta D_i$. By definition, we can derive that:

$$1 - \kappa \leq \frac{Q(t_{i1} + \Delta D_i) - Q(\Delta D_i)}{Q(t_{i1} + O_i) - Q(O_i)}$$

Thus, we obtain:

$$(Q(t_{i1} + O_i) - Q(O_i)) \times (1 - \kappa) \leq Q(t_{i1} + \Delta D_i) - Q(\Delta D_i) \quad (5)$$

Given that the outer loop of the Algorithm 1 runs for K steps, we can apply the aforementioned operation at each step of the process. Consequently, we have:

$$(1 - \kappa)Q(O) = (1 - \kappa)Q(\emptyset) + (1 - \kappa) \sum_{i=1}^K Q(O_i + t_{i1}) - Q(O_i).$$

Since $Q(\emptyset) \geq 0$ and combining Eq. 5, we can conclude:

$$(1 - \kappa)Q(O) \leq Q(\emptyset) + \sum_{i=1}^K Q(t_{i1} + \Delta D_i) - Q(\Delta D_i)$$

Finally, based on lines 6-9 of Algorithm 1 at each step, we have $(Q(t_{i2} + \Delta D_i) - Q(\Delta D_i)) \geq (Q(t_{i1} + \Delta D_i) - Q(\Delta D_i))$. Therefore:

$$\begin{aligned} (1 - \kappa)Q(O) &\leq Q(\emptyset) + \sum_{i=1}^K (Q(t_{i1} + \Delta D_i) - Q(\Delta D_i)) \\ &\leq Q(\emptyset) + \sum_{i=1}^K (Q(t_{i2} + \Delta D_i) - Q(\Delta D_i)) = Q(\Delta D_K). \end{aligned}$$

This means that:

$$(1 - \kappa) \times Q(O) \leq Q(\Delta D_K).$$

Thus, the approximation ratio of algorithm is $1 - \kappa$. $\square$

After considering the scenario where only deletions are made, we will continue to analyze our algorithm’s performance in the context of insertion-only scenarios. Fortunately, we conclude that, when considering only insertions, our algorithm is capable of achieving the optimal solution constituted solely by insertions.

THEOREM 4.4. *In the case of considering only insertions, the above greedy approach can reach an optimal insertion result.*

PROOF. (Sketch). **Base Case** ($K = 1$): When $K = 1$, the algorithm iterates through all possible values and identifies the tuple that maximizes the gain function G for insertion. Therefore, the algorithm is optimal at this initial step. **Inductive Step** ($K \geq 2$): Assume that for $K - 1$, the algorithm achieves optimality. We need to show that under this assumption, the algorithm also achieves optimality for K . Suppose that ΔD_{K-1} achieves the maximal Qerror and inserts $K - 1$ tuples. Let the optimal insertion perturbation strategy select tuple t_a at the K -th step, while our Algorithm 1 selects t_b . According to lines 10-13 of Algorithm 1, the following inequality holds:

$$\sum_{j=1}^M \frac{w_{bj}}{C_D(q_j) + 1} \geq \sum_{j=1}^M \frac{w_{aj}}{C_D(q_j) + 1}$$

This implies that $Q'(t_b + \Delta D_{K-1}) \geq Q'(t_a + \Delta D_{K-1})$. Thus, if that optimality is achieved at step $K - 1$, using lines 10-13 of Algorithm 1 ensures that the algorithm remains optimal for step K . $\square$

Although the two aforementioned theorems are only preliminary considering insert-only or delete-only optimal DSA scenarios. We will now demonstrate, through the following corollaries, that in certain special cases, even when considering both insertions and deletions simultaneously, our algorithm can still achieve good approximation performance. The proof of these corollaries can be found in the Appendix D&E [7].

Corollary 1: If the auditor can pre-determine whether each query in the testing workload is insertion-dominated or deletion-dominated, this mathematically removes the inner maximization layer of the Qerror summation in Eq. 3. Under this condition, ΔD of Algorithm 1 retains the following property: $Q(\Delta D) \geq (1 - \kappa)Q(O)$.

Corollary 2: If the queries meet the following conditions: $M > K$, and for all queries q_j , we have $\text{count}(w_{i,j} \neq 0) \leq 2$ and $C_D(q_j) > K \times M$. Then, Algorithm 1 still possesses the following property: $Q(\Delta D) \geq (1 - \kappa)Q(O)$.

The theorem and corollary demonstrate that while computing the optimal DSA is NP-Hard, Algorithm 1 offers a polynomial-time $(1 - \kappa)$ -approximation strategy under practical constraints. The analysis presented allows us to approximate the worst case of learned estimators within polynomial-time, prior to the occurrence of catastrophic estimation errors. This insight enables database administrators to effectively quantify potential weaknesses in estimators, providing a foundation for robust query optimization strategies.

5 EXPERIMENT

In this section, we provide an experimental analysis of the robustness limits of estimators under near worst-case data drifts. We will use our experiments to answer the following questions:

1. **Robustness Quantification:** Can the proposed DSA techniques effectively evaluate the robustness limits of data-driven, query-driven, and hybrid estimators? Do these estimators exhibit high sensitivity to small, targeted data modifications? (§ 5.3)
2. **Scalability:** How do variations in data size, query scale, and the selection of perturbed tables influence the efficiency and effectiveness of the proposed DSA technique? (§ 5.4)
3. **Consequence & Defense:** How would the degraded estimator mislead the query optimizer into selecting suboptimal query plans? What are the characteristics of such plans? Do we have countermeasures to mitigate the impact of DSA and enhance the worst-case performance for learned estimators? (§ 5.5)

5.1 Experimental Setup

Datasets: We evaluate sensitivity analysis in multi-table settings using two widely adopted benchmarks, IMDB-JOB and STATS-CEB, which are commonly used for multi-table CE evaluation [36, 47, 56, 60, 71, 79]. IMDB-JOB is based on the IMDB database [1] and contains six relations with 14 attributes and 62,118,470 tuples [53]. We use the open-source JOB-light workload [3], which includes 70 queries and 696 subqueries. STATS-CEB is derived from anonymized StackExchange data [2, 36] and contains eight relations with 43 attributes and 1,029,842 tuples. We use the open-source workload [4] with 146 queries and 2,603 subqueries. Following prior protocols [4, 56], we normalize each query (and optimizer sub-expressions) into a unified cardinality-estimation form SELECT

COUNT(*) over the same FROM/JOIN/WHERE clause; thus W_{Test} consists of SPJ subqueries without output aggregates (e.g., MIN) [53]. To train query-driven estimators, we follow Li et al. [56] and generate 2,000 training queries along with their subqueries.

CE models: Based on various studies [56, 60, 71], we selected the following most competitive and representative learned estimators covering the paradigms of data-driven, query-driven, and hybrid: (1) *LWNN (Query-driven)* [30]: uses MLPs to map query representations to cardinalities. We follow the instructions in [36] to extend LWNN to support joins. (2) *MSCN (Query-driven)* [48]: the most famous learned query-driven method based on a multiset convolutional network model. (3) *NeuroCard (Data-driven)* [76]: learns a single deep auto-regressive model for the joint distribution of all tables in the database and estimates the cardinalities using the learned distribution. Following the settings in [76], we set the sampling size of the NeuroCard to 8,000. (4) *FactorJoin (Data-driven)* [71]: integrates classical join-histogram methods and learned Bayes Network into a factor graph. (5) *ALECE (Hybrid)* [56]: uses a transformer to learn the mapping from query representation and histogram features to cardinality. (6) *RobustMSCN (Hybrid)* [60]: is an improvement to the basic MSCN method [48], aims at improving the robustness of the naive MSCN by employing masked training and incorporating data-level features such as PG estimates and join bitmaps. (7) *Oracle:* is the oracle E_O which is capable of utilizing all information from the contaminated database D' to obtain query q 's true cardinality $C_{D'}(q)$ within D' .

Baselines: We compare DSA with the estimators before any perturbations (Clean). Additionally, we compare our approach against the following baselines, selected for their relevance in reducing performance overheads in learned estimators or their application in analogous NP-Hard database optimization tasks [14, 41, 56, 60, 81]: (1) *PACE* [81]: is the state-of-the-art query-centric analysis method on learned query-driven estimators. PACE tests query-driven estimators by contaminating historical workloads.

- (2) *Random Generation (Random)*: randomly selects tuples from the training database to delete or duplicate.
- (3) *Sequential Sort (Sequential)*: Simulates natural temporal drift by modifying data sorted by primary key, adopting the standard sequential update strategy used in robustness analysis [56, 60].
- (4) *Greedy (Heuristic)*: Greedily removes tuples with the highest joint weights to investigate the fragility from “heavy-hitter” data.
- (5) *Simulated Annealing (SA)* [15]: constructs perturbation strategies via a probabilistic hill-climbing algorithm and maximizes Eq. 3.
- (6) *Genetic Algorithm (GA)* [59]: treats a set of potential perturbation strategies as a gene, by randomly selecting several sets of possible strategies and using them as a population for genetic crossover and mutation, therefore maximizing Eq. 3.

Experimental Settings: *Estimators Settings:* For the cardinality estimators to be tested, we train them on the contaminated database D' and test them on the clean database D . This choice intentionally isolates the estimator-induced errors in plan selection from confounding factors such as changed table sizes or changed join selectivities at execution time: the runtime and true cardinalities are always measured on the same clean database D , while only the estimator's internal “belief” is shifted by training on D' . That is to say, the cardinality labels for the Qerror evaluation and the data used to

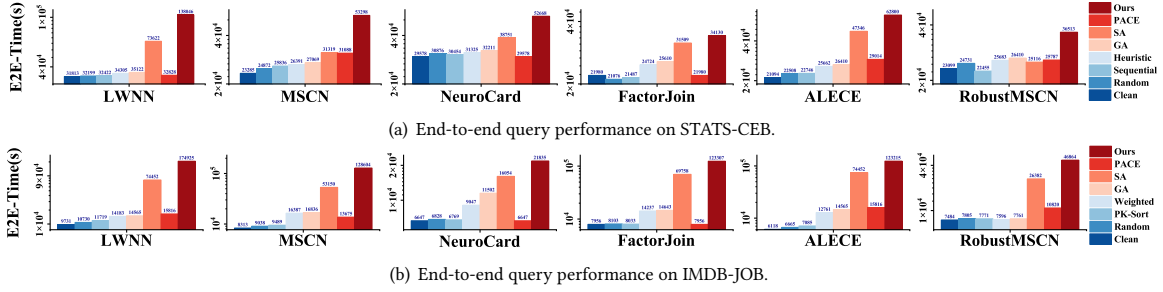


Figure 4: End-to-end (E2E) latency degradation per query on STATS/IMDB database.

evaluate the actual latency are both derived from the clean database D . For hybrid methods, we provide the data features of the clean database during testing. Specifically, we supply ALECE with histogram representations from the clean database D , and provide RobustMSCN with PG estimates and joint sampling features from the clean database. **Stress Test Settings:** In § 5.3, we target the `title` table of the IMDB-JOB database and the `posts` table of the STATS-CEB database. Guided by the budget calibration in § 5.2, we fix the maximum perturbation budget at 20% of the targeted table. We emphasize that this threshold is selected to ensure the sufficient “activation” of baseline methods for effective performance stratification, rather than a necessity for DSA, which exposes combinatorial vulnerabilities at much lower budgets (e.g., 1.25% in table 1). Consequently, our setting circumvents the massive global drifts ($> 20\%$) typically required by prior studies (e.g., ALECE [56]), achieving rigorous stress testing with a global modification scale that is an order of magnitude lower. Specifically, the auditor modifies only $\frac{5 \times 10^5}{6.2 \times 10^7} \approx 0.8\%$ of the total records in IMDB-JOB, and $\frac{1.8 \times 10^4}{1 \times 10^6} \approx 1.2\%$ in STATS-CEB. **Metrics:** We use the metrics defined in § 2.3 to evaluate the stress test’s effectiveness. We use Qerror to measure the degradation in estimation accuracy to evaluate the quality of the generated query plans, and end-to-end (E2E) time to assess the impact of estimation on the end-to-end execution of queries. We utilize the framework developed in [36] to inject the cardinalities predicted by the estimator into Postgres and evaluate the end-to-end execution time.

Environments: Our end-to-end experiments were conducted on an individual Huawei Cloud server with 4 Intel(R) Xeon(R) Platinum 8378A CPUs and 32GB RAM and 1T SSD. Apart from that, all remaining experiments were run on a server with 4 RTX A6000 GPUs, 40 Intel(R) Xeon(R) Silver 4210R CPUs, and 504GB RAM.

5.2 Robustness Sensitivity & Budget Calibration

Our primary objective is to calibrate a minimal perturbation budget that can stratify baseline performance, enabling fair comparison by ensuring all drift generation methods are sufficiently “activated.” To isolate pure distribution shift effects from model-specific biases, we use the oracle estimator (E_O) as a probe. By granting E_O full knowledge of the perturbed database D' , any performance degradation on the current database D can be attributed solely to the distribution shift ΔD , excluding confounding factors such as binning errors or neural approximation limits. Under this controlled setting, we test DSA against two approaches on STATS (with extended analyses in Appendix J [7]), yielding two key insights:

First, robustness analysis under limited budgets is inherently a combinatorial problem. As shown in Table 1, with a negligible

budget of 1.25% of the target table (only 0.11% of the total database), heuristic baselines fail to induce meaningful degradation (Median Qerror ≈ 1.06), challenging the assumption that removing “heavy-hitters” alone suffices. In contrast, DSA exposes structural fragility by identifying a worst-case subset that drives the 95th percentile error to 396 and the maximum error to 1.64×10^6 under the same budget. Second, fair comparison requires a budget calibrated to “activate” baseline methods. Contrary to prior work suggesting that large global drifts ($> 20\%$ of the *entire* database) are necessary for performance decay [56, 60], our results show that baselines already exhibit observable degradation when 20% of the target table is modified ($\approx 1.79\%$ of the total database), where the Heuristic approach reaches a Median Qerror of 3.20. Although DSA saturates at the much lower 0.11% global level, we standardize the perturbation budget at this 20% target-level threshold ($\approx 1.79\%$ global) for subsequent experiments in Section 5.3, ensuring both a rigorous evaluation space for heuristic methods and evidence that worst-case analysis does not require massive global drifts.

Takeaways: DSA reaches CE boundaries at negligible data drifts (0.11% of total DB) where heuristics remain silent. For fair comparison, we adopt a conservative 20% budget ($\approx 1.79\%$ of total DB) to activate baselines, enabling effective performance stratification without massive updates.

 Table 1: Budget Calibration on STATS-CEB via E_O .

Drift Constraint / Budget			Audit Method	Sensitivity Metric (Qerror)			
Rows	Table %	Total DB %		50%	90%	95%	Max
1,150	1.25%	0.11%	Sequential	1.00	1.04	1.06	1.15
			Heuristic	1.06	1.67	2.74	58.4
			DSA (Ours)	1.40	16.5	396	1.64×10^6
2,299	2.50%	0.22%	Sequential	1.02	1.09	1.15	1.25
			Heuristic	1.10	2.11	3.34	78.7
			DSA (Ours)	1.55	146	4.19×10^3	1.64×10^6
4,599	5.00%	0.45%	Sequential	1.05	1.23	1.43	1.94
			Heuristic	1.27	5.65	9.17	137
			DSA (Ours)	1.81	1.62×10^6	1.02×10^8	1.11×10^9
9,198	10.0%	0.89%	Sequential	1.13	1.38	1.66	2.24
			Heuristic	1.94	15.0	22.6	1.39×10^3
			DSA (Ours)	9.82	4.15×10^7	1.47×10^8	1.78×10^{10}
18,395	20.0%	1.79%	Sequential	1.38	1.97	2.25	4.09
			Heuristic	3.20	56.2	110	4.75×10^3
			DSA (Ours)	1.67×10^4	1.00×10^8	2.50×10^8	1.78×10^{10}

5.3 Decline of the Estimator’s Performance

Qerror Distribution Analysis: We evaluate estimator robustness across the Qerror distribution (Table 2), as the median reflects general-purpose utility [77] while tail errors govern the risk of catastrophic query optimization [78, 81]. Our approach consistently dominates all baselines across query-driven, data-driven, and hybrid architectures. For query-driven models (e.g., LWNN, MSCN),

Table 2: Percentile Qerrors on STATS-CEB and IMDB-JOB.

PARADIGM	CE METHOD	PERTURBATION METHOD	IMDB-JOB QERROR					STATS-CEB QERROR				
			50%	90%	95%	99%	MAX	50%	90%	95%	99%	MAX
Query-Driven	LWNN	Clean	2.661	65.04	117.6	1567	$5.67 \cdot 10^4$	4.762	25.98	47.18	1182	$2.66 \cdot 10^4$
		Random	2.271	64.52	118.2	1595	$3.81 \cdot 10^4$	4.489	20.37	32.10	1505	$3.51 \cdot 10^4$
		Sequential	3.675	81.12	228.2	9489	$3.84 \cdot 10^4$	5.085	44.12	143.0	1416	$3.86 \cdot 10^4$
		Heuristic	13.9	3183	$1.40 \cdot 10^4$	$1.40 \cdot 10^5$	$2.57 \cdot 10^5$	5.44	85	601	$3.73 \cdot 10^4$	$4.03 \cdot 10^5$
		GA	48.61	1105	5570	$8.41 \cdot 10^4$	$9.60 \cdot 10^4$	21.84	377.7	1167	9173	$5.52 \cdot 10^4$
		SA	184.1	$2.05 \cdot 10^4$	$4.21 \cdot 10^4$	$1.28 \cdot 10^5$	$1.91 \cdot 10^5$	39.87	1065	3847	$4.65 \cdot 10^4$	$3.67 \cdot 10^6$
		PACE	27.54	1359	6542	$4.79 \cdot 10^4$	$9.59 \cdot 10^4$	26.44	654.8	2486	$1.36 \cdot 10^4$	$7.23 \cdot 10^7$
		Ours	350.3	$3.29 \cdot 10^4$	$9.12 \cdot 10^4$	$1.51 \cdot 10^5$	$9.35 \cdot 10^8$	532.8	$2.43 \cdot 10^4$	$8.35 \cdot 10^4$	$1.14 \cdot 10^6$	$1.14 \cdot 10^8$
	MSCN	Clean	2.076	13.92	53.68	199.3	1653	2.235	99.10	179.1	505.1	2689
		Random	2.085	20.14	59.14	178.1	2116	2.724	161.8	298.8	732.2	4342
		Sequential	3.106	28.77	104.5	494	$1.92 \cdot 10^5$	3.60	180.3	410.5	4610	9812
		Heuristic	9.55	402	2641	$5.05 \cdot 10^4$	$1.40 \cdot 10^5$	4.32	280	355	$2.28 \cdot 10^4$	$6.88 \cdot 10^4$
		GA	18.58	443.2	2111	$2.13 \cdot 10^4$	$1.12 \cdot 10^6$	19.71	397.2	1257	$1.09 \cdot 10^4$	$5.03 \cdot 10^4$
		SA	30.07	2198	6911	$2.51 \cdot 10^4$	$1.71 \cdot 10^5$	38.31	3731	$1.58 \cdot 10^4$	$1.05 \cdot 10^5$	$1.24 \cdot 10^5$
		PACE	54.60	569.1	1437	$4.46 \cdot 10^4$	$2.11 \cdot 10^5$	10.01	117.6	209.2	1488	9724
		Ours	60.23	6201	$3.04 \cdot 10^4$	$1.56 \cdot 10^5$	$1.46 \cdot 10^6$	184.3	$4.63 \cdot 10^6$	$1.16 \cdot 10^7$	$4.74 \cdot 10^7$	$8.08 \cdot 10^8$
Data-Driven	NeuroCard	Clean	1.416	3.312	8.32	18.39	21.51	2.621	1089	6643	$4.95 \cdot 10^4$	$1.71 \cdot 10^6$
		Random	1.869	4.632	8.533	25.97	27.03	2.986	1092	6426	$4.95 \cdot 10^4$	$4.85 \cdot 10^6$
		Sequential	1.876	5.47	9.4	32	43.03	3.18	1351	4326	$8.95 \cdot 10^4$	$2.23 \cdot 10^6$
		Heuristic	4.7	91.8	181	768	4934	3.38	1471	3032	$4.27 \cdot 10^5$	$4.33 \cdot 10^6$
		GA	1.884	11.72	56.56	830.9	971.6	4.09	1073	6690	$6.6 \cdot 10^5$	$1.04 \cdot 10^6$
		SA	39.58	484	1031	3626	6618	3.967	1181	7211	$4.95 \cdot 10^5$	$1.04 \cdot 10^7$
		PACE	1.416	3.312	8.32	18.39	21.51	2.621	1089	6643	$4.95 \cdot 10^4$	$1.71 \cdot 10^6$
		Ours	108.4	2972	3888	$1.53 \cdot 10^4$	$3.47 \cdot 10^4$	15.91	$1.8 \cdot 10^5$	$1.12 \cdot 10^6$	$4.09 \cdot 10^7$	$2.39 \cdot 10^8$
	FactorJoin	Clean	3.015	6.82	39.11	$1.36 \cdot 10^4$	$1.98 \cdot 10^5$	2.018	40.61	78.91	302.7	760.1
		Random	3.176	5.481	45.82	$1.63 \cdot 10^4$	$8.01 \cdot 10^5$	2.072	45.98	64.56	312.98	774.3
		Sequential	3.145	16.21	101.0	$3.25 \cdot 10^4$	$3.64 \cdot 10^5$	2.19	35.7	60.2	455	875
		Heuristic	4.11	312	$3.29 \cdot 10^4$	$8.90 \cdot 10^4$	$1.10 \cdot 10^7$	2.46	218	611	$4.32 \cdot 10^4$	$3.80 \cdot 10^5$
		GA	4.088	156.7	4087	$1.19 \cdot 10^5$	$1.54 \cdot 10^6$	5.03	115.8	296.2	921.8	$1.13 \cdot 10^4$
		SA	3.999	889.6	$1.76 \cdot 10^4$	$8.76 \cdot 10^5$	$3.68 \cdot 10^6$	6.18	784.6	$1.67 \cdot 10^4$	$2.22 \cdot 10^7$	$1.41 \cdot 10^9$
		PACE	3.015	6.82	39.11	$1.36 \cdot 10^4$	$1.98 \cdot 10^5$	2.018	40.61	78.91	302.7	760.1
		Ours	6.706	$1.56 \cdot 10^5$	$2.33 \cdot 10^6$	$4.523 \cdot 10^7$	$1.04 \cdot 10^9$	6.486	2099	$8.05 \cdot 10^4$	$1.72 \cdot 10^8$	$4.52 \cdot 10^9$
Hybrid	ALECE	Clean	1.374	2.611	3.702	16.03	649.4	1.532	3.007	4.321	19.55	48.89
		Random	1.566	2.741	3.499	9.782	439.7	1.648	3.563	4.583	27.04	68.4
		Sequential	1.62	3.77	4.8	12.5	558.6	1.82	4.28	5.84	43.8	92.7
		Heuristic	9.21	153	1169	9480	$8.64 \cdot 10^4$	1.64	56.4	61.2	583	1839
		GA	13.86	159.2	212.9	5547	$1.28 \cdot 10^5$	4.877	76.31	186.9	417.5	2027
		SA	27.01	696.1	2723	$5.06 \cdot 10^4$	$1.49 \cdot 10^5$	4.32	724.5	1405	3458	4631
		PACE	7.229	134.4	265.6	4284	$4.62 \cdot 10^5$	4.08	207.5	544.8	5637	$3.03 \cdot 10^4$
		Ours	29.33	9323	$2.73 \cdot 10^4$	$2.45 \cdot 10^5$	$2.02 \cdot 10^6$	6.442	$3.23 \cdot 10^4$	$2.06 \cdot 10^5$	$5.88 \cdot 10^6$	$1.64 \cdot 10^7$
	RobustMSCN	Clean	1.811	4.415	23.67	355.5	1102	2.288	5.196	7.43	28.62	447.0
		Random	2.145	5.409	23.81	381.5	2547	2.723	5.903	7.743	23.55	4656
		Sequential	2.19	7.45	24.4	406	2213	2.73	7.62	9.97	41.7	7204
		Heuristic	5.03	81.6	113	1348	$2.02 \cdot 10^4$	2.9	26.8	35.4	325	$3.22 \cdot 10^4$
		GA	15.30	109.1	173.8	627.8	$1.83 \cdot 10^4$	3.820	47.62	81.41	274.9	$7.38 \cdot 10^4$
		SA	12.69	186.3	805.5	5705	$2.16 \cdot 10^4$	3.92	83.2	270.5	1225	$2.85 \cdot 10^4$
		PACE	5.019	77.26	278.7	2352	7155	4.84	98.18	426.7	4009	$1.48 \cdot 10^4$
		Ours	15.87	814.7	2185	$1.52 \cdot 10^4$	$2.64 \cdot 10^5$	5.19	2388	8762	$1.32 \cdot 10^5$	$7.06 \cdot 10^5$
Oracle		Clean	1	1	1	1	1	1	1	1	1	1
		Random	1.255	1.294	1.663	1.766	3.338	1.212	1.298	1.319	1.515	2.328
		Sequential	2.15	2.44	2.77	244.8	780.2	1.375	1.969	2.250	3.763	4.092
		Heuristic	18.31	$1.86 \cdot 10^3$	$1.85 \cdot 10^4$	$1.15 \cdot 10^6$	$3.05 \cdot 10^6$	3.200	56.20	110.0	1519	4749
		GA	83.39	230.3	2005	$5.15 \cdot 10^5$	$1.674 \cdot 10^6$	25.97	258.8	1566	8741	$1.44 \cdot 10^4$
		SA	301.1	$2.72 \cdot 10^5$	$5.12 \cdot 10^5$	$1.54 \cdot 10^7$	$3.23 \cdot 10^7$	35.98	2133	4835	$1.43 \cdot 10^4$	$1.56 \cdot 10^6$
		Ours	$1.49 \cdot 10^5$	$1.23 \cdot 10^8$	$5.03 \cdot 10^8$	$3.89 \cdot 10^9$	$9.53 \cdot 10^9$	$1.66 \cdot 10^4$	$1.02 \cdot 10^8$	$2.50 \cdot 10^8$	$2.21 \cdot 10^9$	$7.60 \cdot 10^{10}$

we outperform competitors (SA, PACE, GA, Random) by factors ranging from $6.85\times$ to $1876\times$. Crucially, the stark contrast between our method and the “Random” baseline reveals a fundamental vulnerability: while modern estimators are resilient to stochastic drift, they are strikingly fragile under targeted modifications. This confirms that robustness is distribution-dependent rather than absolute. Specifically, our approach achieves $2.43\times$ – $47.1\times$ degradation at the 50th percentile. This efficacy extends to data-driven models (e.g., NeuroCard), where PACE [81] often fails because it targets historical workloads; in contrast, our strategy perturbs underlying data correlations to degrade even distribution-centric estimators. The most significant impact occurs at the tail (above the 90th percentile), where our method inflates errors by 3–4 orders of magnitude. Such extreme misestimates are fatal for join-order selection, resulting in catastrophic latency. Furthermore, we observe a “perfection paradox”: high-accuracy estimators that closely fit the training oracle E_O (e.g., NeuroCard on IMDB) are structurally vulnerable, suffering median error increases of two orders of magnitude under our stress tests. These results validate the near-optimality of Algorithm 1 (§ 4.3) in navigating the NP-Hard search space and confirm the theoretical transformations in § 3.2.

Decline of the E2E-Time: To assess how different perturbation strategies affect query optimization, we measured end-to-end execution time across multiple estimators. As shown in Figures 4(a) and 4(b), our method consistently produces the worst execution plans and the largest slowdowns. On IMDB-JOB, it increases execution time by $24.2\times$, $2.21\times$, $17.5\times$, and $202\times$ over PACE, SA, GA, and Random, respectively; on STATS-CEB, the increases reach $11.6\times$, $2.34\times$, $10.3\times$, and $41.8\times$. Estimators exhibit varying vulnerability to data-centric sensitivity analysis (LWNN > MSCN > NeuroCard > ALECE > FactorJoin > RobustMSCN), with RobustMSCN showing relative resilience but still slowing by $1.58\times$. Overall, these results confirm that our sensitivity analysis exposes estimator fragility and drives optimizers toward significantly suboptimal plans.

Takeaways: Our DSA framework systematically uncovers vulnerabilities across data-driven, query-driven, and hybrid estimators. Moving beyond simple heuristics, our global optimization identifies strategic data subsets that inflict disproportionate damage. DSA significantly degrades worst-case accuracy, forcing suboptimal execution plans and excessive latency. Notably, even hybrid models offer only certain robustness against such targeted perturbations.

5.4 Scalability Study

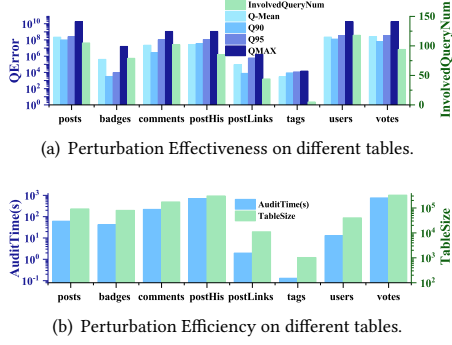


Figure 5: Scalability on different tables.

Scalability on Perturbed Tables: To validate the theoretical perturbation effectiveness on different tables, in Figure 5(a) and Figure 5(b), we conducted experiments by switching the targeted tables within the STATS-CEB database. We selected the oracle E_O as the target. The perturbation budget was set at 20% of the rows in the targeted table. Our findings indicate that, despite the varying number of queries associated with different tables in the STATS-CEB workload, our method can significantly impact E_O for any table, resulting in a maximum Qerror ranging from 10^4 to 10^{10} . Additionally, we observed that the more queries a table encompasses, the greater the impact of the perturbation. For instance, when the number of queries increased from 10 (in tags) to 76 (in the votes), the maximum Qerror escalated from 10^4 to 10^{10} .

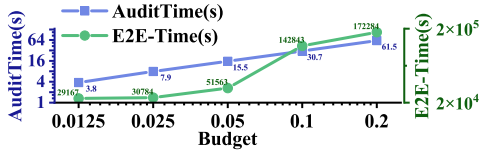


Figure 6: Scalability on different budget.

Scalability w.r.t Perturbation Budget. We evaluate the execution time of DSA on the E_O estimator while increasing the perturbation budget from 1.25% to 20% on the posts table. As illustrated in Figure 6, the audit time grows linearly with the budget size K . Even at the maximum tested budget of 20% (where estimator performance is completely degraded), the analysis completes in acceptable limits.

Takeaways: The overhead of our proposed DSA analysis strategy is linear to the perturbation budget and correlated with table size and query number. Moreover, the more queries that involve the targeted table, the larger the perturbation budget, and consequently, the greater the benefit derived from the sensitivity analysis.

5.5 Utility: Auditing, Guidance, and Diagnostics

In this section, we explore the utility of our DSA. Based on the worst-case discovered by DSA, we devised a spectrum of five distinct approaches to investigate and mitigate worst-case failures. We detail three primary applications in the main text, while extending our analysis to two further mechanisms to boost the worst-case performance in our Appendix (G-I) [7].

Application I: Robust Model Maintenance Auditing: A critical application of DSA is determining when to update learned models

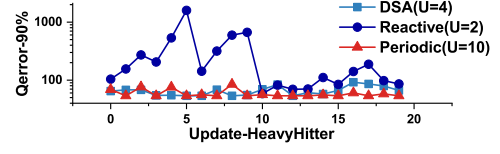


Figure 7: Utility: DSA as an Early Warning Sentinel.

in dynamic environments. In real-world scenarios with "Heavy Hitter" updates, fixed strategies often fail. To simulate this, we utilized the STATS-CEB-post table and the LWNN estimator, performing 20 rounds of updates where each round modifies 1% of the data (detailed settings are provided in the Appendix I [7]). As shown in Figure 7, the Reactive (Post-hoc) strategy triggers retraining only after detecting high errors; consequently, it incurs unavoidable production losses, evidenced by catastrophic Qerror spikes (> 1000) before mitigation occurs. The Periodic (High Budget) strategy ensures stability by blindly retraining every two rounds ($U = 10$) but suffers from prohibitive computational overhead. In contrast, our DSA-Guided Sentinel proactively identifies critical tuple shifts. It achieves stability comparable to the periodic method with only four effective updates ($U = 4$), saving 60% of the budget. This confirms DSA's utility as a cost-effective sentinel for data drift.

Application II: Guidance for Robust System Design Beyond determining maintenance timing, the value of our DSA also lies in guiding the architectural design of more robust estimators. We investigated whether architectural redundancy enhances robustness. As shown in Figure 8, we tested an ensemble strategy guided by sensitivity analysis, which performs a weighted fusion of six different estimators. While this ensemble approach mitigates performance degradation in worst-case scenarios to some extent, significant room for optimization remains. To further explore architectural defenses, we investigate Hardening via Controlled Noise Injection in Appendix H [7]. This strategy establishes a statistical "buffer zone" against distribution shifts and reveals a critical tradeoff between nominal precision and worst-case stability.

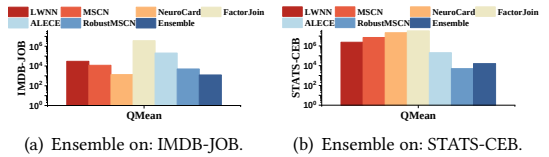


Figure 8: Robustness Assessment of Ensemble Architectures.

Application III: Diagnostics of Optimizer Behavior To understand how estimator worst-case failures propagate to the downstream query optimizer, we utilized DSA to simulate worst-case scenarios on IMDB-JOB and STATS-CEB using LWNN, diagnosing the specific vulnerabilities of the query optimizer (Figure 9).

Vulnerability 1: Join Pattern Sensitivity. Our audit reveals that the optimizer is extremely sensitive to cardinality underestimation, leading to drastic shifts in join operators. As shown in Figure 9, under the pressure of DSA-generated updates, the usage of Nested Loop Joins surged by 128% (IMDB-JOB) and 61% (STATS-CEB), while efficient Hash Joins declined significantly. This indicates that when misled by worst-case estimates, the optimizer regresses to operators suitable only for small data, severely degrading complex analytical queries. A stark example is the regression of IMDB-JOB

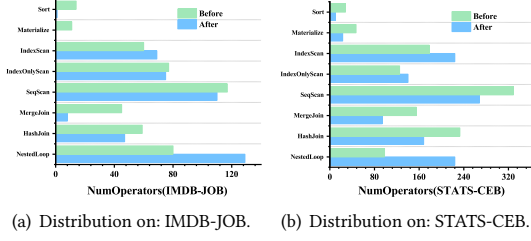


Figure 9: Diagnostic: Operator Distribution Shift.

Q60 (Figure 10(a) vs. Figure 10(b)), where the optimizer abandoned Hash Joins for Index Nested Loops, causing a massive increase in latency. This finding prompts a natural heuristic defense: disabling risky operators. In Appendix G[7], we evaluate this “Safety-First” configuration (disabling NLJs). Our audit reveals that while this heuristic works for average cases, it fails at the tail, leading the optimizer into a “Merge Join Trap” (52× degradation). This persistent fragility underscores the inherent difficulty of the robustness problem, indicating that resolving these estimator blind spots requires structural advancements beyond simple operator constraints, necessitating broader community efforts.

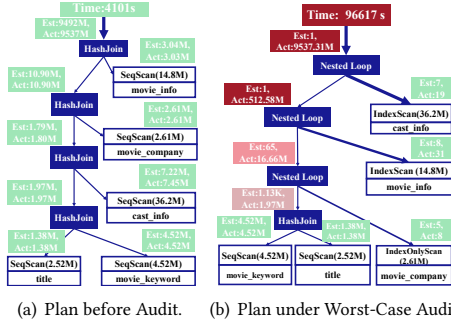


Figure 10: Case Study: Diagnosing Plan Regression on Q60.

Vulnerability 2: Cache Efficiency. The diagnostic also uncovered a hidden cost: reduced cache efficiency. We observed that materialized operators decreased by 90% in IMDB and 51% in STATS. This implies that worst-case errors not only affect current execution but also mislead the global plan into bypassing materialized views, triggering expensive redundant computations.

Takeaways: Our DSA offers five pathways to mitigate worst-case failures. We demonstrate: (1) **Proactive Maintenance** (saving 60% budget via “Sentinel” updates); (2) **Architectural Benchmarking** (testing ensembles); and (3) **Root Cause Diagnostics** (tracing optimizer faults). We further explore (4) **Operator Restrictions** and (5) **Noise Injection** defenses in the Appendix (G-I) [7].

6 RELATED WORK

Learned Cardinality Estimators: Modern cardinality estimators can be categorized into three paradigms: query-driven [30, 48, 70], data-driven [57, 71, 76, 77, 77] and hybrid [56, 60]. (1) Query-driven estimators employ neural networks [30, 48, 70] to learn mappings

from query representations to true cardinalities. Although query-driven methods can achieve quick and lightweight estimations, it is confirmed by many studies that query-driven estimators suffer from changes in testing query distribution, due to data drifts or shifts within query workloads [56, 66, 67, 77]. (2) Data-driven estimators learn joint data distributions and sample from deep models [47, 76, 77] to estimate cardinalities. They are robust against workload drifts [67, 77]. (3) Hybrid estimators [37, 48, 56, 60, 60, 70] combine data statistics with query features for unified learning. Commonly used data features include histograms [56, 79], sample collections [48, 60], and estimations from PostgreSQL [37, 60, 79]. **Robustness of Learned Database Components:** Nowadays, the robustness and stability of learned database components have garnered widespread attention [75, 75, 81, 84]. Kornaropoulos et al. [75, 75] introduced algorithmic complexity analysis targeting learned indexes by constructing adversarial inputs that cause learned indexes to fail. Additionally, Zheng et al. [84] proposed robustness evaluation aimed at learning-based index advisors. The work most closely related to ours is PACE proposed by Zhang et al. [81], which stress-tested query-driven estimators by adding noise to their training workloads. However, PACE has a clear limitation: it can only target query-driven models. In contrast, the sensitivity analysis framework proposed in this paper can evaluate the robustness limits of nearly all types of cardinality estimators.

Adversarial Data Drifts in ML: Traditional ML adversarial robustness is typically analyzed under the white-box assumption, where the auditor has full knowledge of the internal model [13, 17, 18, 42, 73, 74, 82]. For example: (1). Under the assumption of linear regression, Jagielski et al. [13] proposed an optimization framework for analyzing the sensitivity of linear regression models. (2). For support vector machines (SVMs) [17, 73, 74], adversarially crafted training data can manipulate the decision boundary, thereby increasing the error rate. (3). In the context of neural networks, Yang et al. [82] introduced a gradient-based technique to generate worst-case inputs. In contrast, our work focuses on a more realistic black-box scenario, where the auditor lacks knowledge of the internal model details. Specifically, the auditor is unaware whether the internal model is based on neural networks [56, 76], decision trees [30, 38, 88], Bayesian networks [71, 72], or other emerging models [66, 79].

7 CONCLUSION

This work provides the first theoretical investigation into the stability limits of learned cardinality estimators under minimal, targeted data drifts. We introduce a black-box Data-centric Sensitivity Analysis (DSA) framework, proving that while finding the optimal worst-case drift is NP-Hard, a polynomial-time $(1 - \kappa)$ -approximation is achievable. Experiments on STATS-CEB and IMDB-JOB benchmarks show our stress test’s effectiveness: modifying merely 0.8% of database tuples causes a 1000× increase in 90-th percentile Q_{err} and up to 20× longer processing times. These findings expose the fragility of current estimators in real-world scenarios. We also propose practical countermeasures against such worst-case drifts, providing valuable insights for developing robust learned optimizers. Future work could focus on creating more resilient estimation techniques that can withstand adversarial data distributions while maintaining reliable performance.

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Due to space limitations, we present more details in this appendix.

APPENDIX ROADMAP

Due to space limitations in the main content, we present supplementary materials in this appendix, organized as follows:

- **Theoretical Proofs (Section A – E):** We provide the complete mathematical derivations and proofs to support our theoretical claims. This includes the NP-hardness of DSA (Theorem 3.4), the detailed derivation of the objective function (Theorem 4.1), the modularity property (Theorem 4.2), and the approximation guarantees for our algorithm (Corollary 1 & 2).
- **Experimental Settings (Section F):** To ensure reproducibility, we detail the specific hardware environments and the non-default PostgreSQL configurations used in our evaluation.
- **Extended Utility Analysis (Section G – I):** We present further investigations into system robustness, including an analysis of heuristic defenses (the “Merge Join Trap”), the potential of noise injection for hardening estimators, and detailed settings for the heavy-hitter maintenance audit.
- **Extended Budget Analysis (Section J):** We present further sensitivity analysis on IMDB.
- **Oracle-to-Learned Transfer Analysis (Section K):** We provide the formal theoretical justification for using the oracle-based objective as a proxy for learned estimator reliability. We prove that drift-induced errors in learned models are mathematically bounded by the oracle discrepancy (Theorem K.7) and demonstrate that standard log-space training objectives directly control this transfer bound (Theorem K.8).
- **Real-world Applicability & Partial Drifts (Section L):** We extend our discussion regarding the strictness of worst-case analysis.

A PROOF OF THEOREM 3.4

We present the proof of Theorem 3.4 here:

PROOF. We aim to prove that when $x > K \times M$, the benefit in Q_{error} resulting from inserting I tuples ($2 \leq I \leq K$) into the database constructed from Theorem 3.1 is smaller than the benefit obtained by not deleting any two tuples corresponding to any q_x . Consequently, the plan of inserting I tuples can be disregarded and replaced with I delete operations. And we can therefore reuse the PTIME reduction in theorem 3.3, construct a polynomial-time reduction from the DKS problem to the DSA problem when simultaneously considering insertions and deletions. Therefore the DSA problem considering both insertions and deletions remains NP-Hard.

For a fixed query q_x , deleting its two tuples in R_x yields a benefit $g_{\text{Del}} = 2x + 1$. Given that $x > K \times M$, we have

$$g_{\text{Del}} > 2 \times K \times M + 1.$$

Assume that all M queries share one common test tuple in R_x . Therefore, repeatedly inserting into this special tuple would be the optimal solution for inserting I tuples, which provides an upper

bound on the Q_{error} when inserting I tuples. That is,

$$g_{\text{Ins}} \leq M \times \frac{I + 2}{2}.$$

Given that $g_{\text{Del}} > 2 \times K \times M + 1$ and $K \geq I$, we have

$$g_{\text{Del}} > 2 \times K \times M > 2 \times I \times M = \frac{IM}{2} + \frac{3IM}{2}.$$

Since $I \geq 2$, we have $\frac{3IM}{2} > M$, and thus

$$2 \times I \times M = \frac{IM}{2} + \frac{3IM}{2} > \frac{IM}{2} + M \geq g_{\text{Ins}}.$$

In conclusion, we deduce that the benefit g_{Del} obtained by deleting any two tuples corresponding to query q_x in R_x is greater than the benefit g_{Ins} of inserting I tuples ($2 \leq I \leq K$). $\square$

B PROOF OF THE DERIVATION IN THEOREM 4.1

Given that $Q_j(X) = \frac{C_D(q_j)+1}{C_D(q_j)+1+\sum_{t_i \in X} t_i \cdot \beta \times w_{ij}}$ and

$$w'_{ij} = \frac{w_{ij}}{1+C_D(q_j)}, \quad S_j(X) = \sum_{t_i \in X} t_i \cdot \beta \times w'_{ij}$$

We prove that the expression

$$Q_j(A \cup B) + Q_j(A \cap B) - Q_j(A) - Q_j(B)$$

can be reorganized in the following form:

$$\frac{(2 + S_j(A) + S_j(B))(S_j(A \setminus B))(S_j(B \setminus A))}{(1 + S_j(A \cup B))(1 + S_j(A \cap B))(1 + S_j(A))(1 + S_j(B))}$$

PROOF. Based on the weights w'_{ij} and the function $S_j(X)$, we define the function $Q_j(X)$ as follows:

$$Q_j(X) = \frac{C_D(q_j) + 1}{C_D(q_j) + 1 + \sum_{t_i \in X} t_i \cdot \beta \times w_{ij}} = \frac{1}{1 + S_j(X)}.$$

Therefore, for any sets A and B , the following relationship holds:

$$\begin{aligned} & Q_j(A \cup B) + Q_j(A \cap B) - Q_j(A) - Q_j(B) \\ &= \frac{1}{1 + S_j(A \cup B)} + \frac{1}{1 + S_j(A \cap B)} - \frac{1}{1 + S_j(A)} - \frac{1}{1 + S_j(B)} \end{aligned}$$

We examine the first two terms of the above summation:

$$\frac{1}{1 + S_j(A \cup B)} + \frac{1}{1 + S_j(A \cap B)}$$

This can be combined as follows:

$$\begin{aligned} & \frac{1}{1 + S_j(A \cup B)} + \frac{1}{1 + S_j(A \cap B)} \\ &= \frac{(2 + S_j(A \cup B) + S_j(A \cap B)) \times (1 + S_j(A)) \times (1 + S_j(B))}{(1 + S_j(A \cup B))(1 + S_j(A \cap B))(1 + S_j(A))(1 + S_j(B))} \\ &= \frac{2 + S_j(A \cup B) + S_j(A \cap B)}{(1 + S_j(A \cup B))(1 + S_j(A \cap B))} \times \frac{(1 + S_j(A))(1 + S_j(B))}{(1 + S_j(A))(1 + S_j(B))} \end{aligned}$$

Similarly, the last two terms of the summation are:

$$\frac{1}{1 + S_j(A)} + \frac{1}{1 + S_j(B)}$$

These terms can be combined as:

$$\begin{aligned}
 & \frac{1}{1+S_j(A)} + \frac{1}{1+S_j(B)} \\
 &= \frac{(2+S_j(A)+S_j(B)) \times (1+S_j(A \cup B)) \times (1+S_j(A \cap B))}{(1+S_j(A))(1+S_j(B))(1+S_j(A \cup B))(1+S_j(A \cap B))} \\
 &= \frac{2+S_j(A)+S_j(B)}{(1+S_j(A))(1+S_j(B))} \times \frac{(1+S_j(A \cup B))(1+S_j(A \cap B))}{(1+S_j(A \cup B))(1+S_j(A \cap B))}
 \end{aligned}$$

Notably, we observe that:

$$2+S_j(A)+S_j(B) = 2+S_j(A \setminus B)+2S_j(A \cap B)+S_j(B \setminus A) = 2+S_j(A \cup B)+S_j(A \cap B)$$

Therefore, by combining the above results, we obtain:

$$\begin{aligned}
 & \frac{1}{1+S_j(A \cup B)} + \frac{1}{1+S_j(A \cap B)} - \frac{1}{1+S_j(A)} - \frac{1}{1+S_j(B)} = \\
 & \frac{(2+S_j(A)+S_j(B))((1+S_j(A))(1+S_j(B)) - (1+S_j(A \cup B))(1+S_j(A \cap B)))}{(1+S_j(A \cup B))(1+S_j(A \cap B))(1+S_j(A))(1+S_j(B))}
 \end{aligned}$$

Additionally, we observe that:

$$\begin{aligned}
 & (1+S_j(A))(1+S_j(B)) - (1+S_j(A \cup B))(1+S_j(A \cap B)) \\
 &= [1+S_j(A \cap B)+S_j(A \setminus B)] [1+S_j(A \cap B)+S_j(B \setminus A)] \\
 & \quad - [1+S_j(A \cap B)+S_j(A \setminus B)+S_j(B \setminus A)] (1+S_j(A \cap B)) \\
 &= S_j(A \setminus B) \times S_j(B \setminus A).
 \end{aligned}$$

Therefore, we can substitute the equation in and have:

$$\begin{aligned}
 & \frac{1}{1+S_j(A \cup B)} + \frac{1}{1+S_j(A \cap B)} - \frac{1}{1+S_j(A)} - \frac{1}{1+S_j(B)} \\
 &= \frac{(2+S_j(A)+S_j(B))(S_j(A \setminus B))(S_j(B \setminus A))}{(1+S_j(A \cup B))(1+S_j(A \cap B))(1+S_j(A))(1+S_j(B))}.
 \end{aligned}$$

Therefore we proofed that:

$$\begin{aligned}
 & \mathbb{Q}'(A \cup B) + \mathbb{Q}'(A \cap B) - \mathbb{Q}'(A) - \mathbb{Q}'(B) \\
 &= \frac{(2+S_j(A)+S_j(B))(S_j(A \setminus B))(S_j(B \setminus A))}{(1+S_j(A \cup B))(1+S_j(A \cap B))(1+S_j(A))(1+S_j(B))}.
 \end{aligned}$$

□

C PROOF OF THEOREM 4.2

We present the proof of Theorem 4.2 here:

PROOF. We aim to demonstrate that the objective

$$\mathbb{Q}'(X) = \sum_{j=1}^M \mathbb{Q}_j(X) = \sum_{j=1}^M \frac{C_D(\mathbf{q}_j) + 1 + \sum_{t_i \in X} t_i \cdot \beta \times w_{ij}}{C_D(\mathbf{q}_j) + 1}$$

Due to the linear nature of the target $\mathbb{Q}'(X)$, we can move the summation over $t_i \cdot \beta$ from the inner layer to the outer layer, rearrange $t_i \cdot \beta$, and combine elements based on the sets $(A \setminus B)$, $(B \setminus A)$, and $(A \cap B)$. Then, we prove that $\mathbb{Q}'(X)$ satisfies the modular property.

We can rewrite $\mathbb{Q}'(X)$ as:

$$\mathbb{Q}'(X) = M + \sum_{j=1}^M \frac{\sum_{t_i \in X} t_i \cdot \beta \times w_{ij}}{C_D(\mathbf{q}_j) + 1}.$$

Let us define $H_i = \sum_{j=1}^M \frac{w_{ij}}{C_D(\mathbf{q}_j) + 1}$. Consequently, we have

$$\mathbb{Q}'(X) = M + \sum_{i=1}^N t_i \cdot \beta \times H_i.$$

Considering the left-hand side, we obtain:

$$\mathbb{Q}'(A \cup B) + \mathbb{Q}'(A \cap B) = \sum_{t_k \in A \cup B} t_k \cdot \beta \times H_k + \sum_{t_k \in A \cap B} t_k \cdot \beta \times H_k.$$

This can be expanded as:

$$\sum_{t_k \in A \setminus B} t_k \cdot \beta \times H_k + 2 \sum_{t_k \in A \cap B} t_k \cdot \beta \times H_k + \sum_{t_k \in B \setminus A} t_k \cdot \beta \times H_k.$$

On the other hand, the right-hand side is:

$$\mathbb{Q}'(A) + \mathbb{Q}'(B) = \sum_{t_k \in A} t_k \cdot \beta \times H_k + \sum_{t_k \in B} t_k \cdot \beta \times H_k.$$

$$\sum_{t_k \in A \setminus B} t_k \cdot \beta \times H_k + 2 \sum_{t_k \in A \cap B} t_k \cdot \beta \times H_k + \sum_{t_k \in B \setminus A} t_k \cdot \beta \times H_k.$$

Therefore, we have

$$\mathbb{Q}'(A) + \mathbb{Q}'(B) = \mathbb{Q}'(A \cup B) + \mathbb{Q}'(A \cap B).$$

This equality demonstrates the modular property. □

D PROOF OF COROLLARY 1

Here, we will prove our Corollary 1. When the auditor specifies which queries to insert and which to delete, our optimization goal becomes

$$\begin{aligned}
 \text{Max} : & \sum_{\mathbf{q}_j \in \text{Insert}} \frac{C_D(\mathbf{q}_j) + 1 + \sum_{i=1}^N t_i \cdot \beta \times w_{ij}}{C_D(\mathbf{q}_j) + 1} \\
 & + \sum_{\mathbf{q}_j \in \text{Delete}} \frac{C_D(\mathbf{q}_j) + 1}{C_D(\mathbf{q}_j) + 1 + \sum_{i=1}^N t_i \cdot \beta \times w_{ij}} \\
 \text{s.t.} & \sum_{i=1}^N |t_i \cdot \beta| \leq K, \quad t_i \cdot \beta \in \{-1, 0, 1, \dots, K\}
 \end{aligned}$$

According to Theorem 4.1 and Theorem 4.2, the first term of the optimization objective

$$\sum_{\mathbf{q}_j \in \text{Insert}} \frac{C_D(\mathbf{q}_j) + 1 + \sum_{i=1}^N t_i \cdot \beta \times w_{ij}}{C_D(\mathbf{q}_j) + 1}, \quad (6)$$

exhibits modular properties. In contrast, the second term of the optimization objective

$$\sum_{\mathbf{q}_j \in \text{Delete}} \frac{C_D(\mathbf{q}_j) + 1}{C_D(\mathbf{q}_j) + 1 + \sum_{i=1}^N t_i \cdot \beta \times w_{ij}}, \quad (7)$$

demonstrates supermodular characteristics. The combination exhibits supermodular characteristics. This aligns with the conditions outlined in [12], where the objective is defined as the sum of a supermodular function and a submodular function with zero curvature. Furthermore, Bai et al. [12] have demonstrated that this linear combination does not impact the approximation ratio of the

greedy algorithm when applied to such linear summation functions. As a result, the approximation ratio remains:

$$\mathbb{Q}(\Delta D) \geq (1 - \kappa)\mathbb{Q}(O). \quad (8)$$

E PROOF OF COROLLARY 2

We aim to demonstrate the following:

1. Under the given conditions, the optimal solution does not include insertions.

Consider the scenario where insertions are contemplated. The upper bound of the potential gain from insertions arises from the repeated insertion across M queries, specifically denoted as $K \times M$. This gain is inferior compared to the cost incurred by deleting one or two tuples corresponding to a query q_x , represented as $C_D(q_j)$. Mathematically, this relationship can be expressed as:

$$K \times M < C_D(q_j)$$

Consequently, the optimal solution favors deletion over insertion, implying that the optimal solution does not incorporate insertions.

2. Under the given conditions, Algorithm 1 will not choose to insert

Algorithm 1 evaluates the remaining undeleted tuples at each step. Consider the L -th step of the algorithm, where we examine an undeleted tuple t_i . We use D_L to represent the current database state and let S_1 denote the set of queries q_j involving t_i with $\text{count}(w_{i,j} \neq 0) = 1$, S_2 denote the set of queries q_j involving t_i with $\text{count}(w_{i,j} \neq 0) = 2$. The potential gain from repeatedly inserting tuples associated with queries in S is given by:

$$\text{gainI} = \sum_{q_j \in S_1} \frac{2 \cdot C_{D_L}(q_j) + 1}{C_{D_L}(q_j) + 1} + \sum_{q_j \in S_2} \frac{C_{D_L}(q_j) + 1 + w_{i,j}}{C_{D_L}(q_j) + 1}$$

On the other hand, if tuple t_i is deleted, the corresponding gain is:

$$\text{gainD} = \sum_{q_j \in S_1} \frac{C_{D_L}(q_j) + 1}{1} + \sum_{q_j \in S_2} \frac{C_{D_L}(q_j) + 1}{C_{D_L}(q_j) + 1 - w_{i,j}}$$

Note that:

$$\frac{C_{D_L}(q_j) + 1}{C_{D_L}(q_j) + 1 - w_{i,j}} - \frac{C_{D_L}(q_j) + 1 + w_{i,j}}{C_{D_L}(q_j) + 1} = \frac{w_{i,j}}{C_{D_L}(q_j) + 1 - w_{i,j}} - \frac{w_{i,j}}{C_{D_L}(q_j) + 1} > 0$$

and:

$$\frac{C_{D_L}(q_j) + 1}{1} - \frac{2 \cdot C_{D_L}(q_j) + 1}{C_{D_L}(q_j) + 1} = C_{D_L}(q_j) - \frac{C_{D_L}(q_j)}{C_{D_L}(q_j) + 1} \geq 0$$

We can compute the difference between the inner summations of the two groups separately, yielding:

$$\begin{aligned} & \sum_{q_j \in S_1} \frac{C_{D_L}(q_j) + 1}{1} - \sum_{q_j \in S_1} \frac{2 \cdot C_{D_L}(q_j) + 1}{C_{D_L}(q_j) + 1} \\ &= \sum_{q_j \in S_1} \left(\frac{C_{D_L}(q_j) + 1}{1} - \frac{2 \cdot C_{D_L}(q_j) + 1}{C_{D_L}(q_j) + 1} \right) > 0 \end{aligned}$$

Additionally, for the second group, we have:

$$\begin{aligned} & \sum_{q_j \in S_2} \frac{C_{D_L}(q_j) + 1}{C_{D_L}(q_j) + 1 - w_{i,j}} - \sum_{q_j \in S_2} \frac{C_{D_L}(q_j) + 1 + w_{i,j}}{C_{D_L}(q_j) + 1} \\ &= \sum_{q_j \in S_2} \left(\frac{C_{D_L}(q_j) + 1}{C_{D_L}(q_j) + 1 - w_{i,j}} - \frac{C_{D_L}(q_j) + 1 + w_{i,j}}{C_{D_L}(q_j) + 1} \right) > 0 \end{aligned}$$

Combining the results from both groups, we obtain:

$$\text{gainI} < \text{gainD}$$

This inequality indicates that the gain from insertion (gainI) is less than the gain from deletion (gainD). Therefore, Algorithm 1 will opt for deletion over insertion at each step. As a result, the algorithm inherently avoids insertions, effectively reducing the problem to an insert-only scenario under the given conditions.

$$\mathbb{Q}(\Delta D) \geq (1 - \kappa)\mathbb{Q}(O). \quad (9)$$

F DETAILED EXPERIMENTAL SETTINGS

To ensure the reproducibility of our results and to provide a fair comparison against state-of-the-art baselines, we provide a comprehensive disclosure of our hardware infrastructure and the specific database configurations used in our end-to-end evaluation.

F.1 Hardware Environments

We decoupled the model training phase from the query execution phase to optimize resource utilization appropriate for each task. The hardware specifications are detailed as follows:

- **End-to-End Query Evaluation Platform:** All end-to-end latency experiments, including the integration of learned estimators into the PostgreSQL kernel, were conducted on an individual Huawei Cloud instance. This environment features **4 Intel(R) Xeon(R) Platinum 8378A CPUs, 32GB of RAM, and a 1TB SSD**. This configuration was chosen to simulate a realistic, resource-constrained database server environment where I/O and CPU efficiency are critical.
- **Model Training Platform:** The training of deep learning-based cardinality estimators (and other baselines requiring GPU acceleration) was performed on a high-performance computing server equipped with **4 NVIDIA RTX A6000 GPUs, 40 Intel(R) Xeon(R) Silver 4210R CPUs, and 504GB of RAM**.

F.2 Software and Database Configuration

For the database system, we utilized **PostgreSQL 13.1**. To maintain strict objectivity and facilitate direct comparisons with prior work, our experimental environment is derived from the standard docker image provided by the *Cardinality Estimation Benchmark* (CEB). Using this standardized container ensures that our operating system dependencies and libraries are consistent with established benchmarks in the community.

F.2.1 PostgreSQL Parameter Settings. We report the specific PostgreSQL configurations used during the runtime evaluation in Table 3. Note that we list only the parameters that differ from the default PostgreSQL configuration (mode: changed).

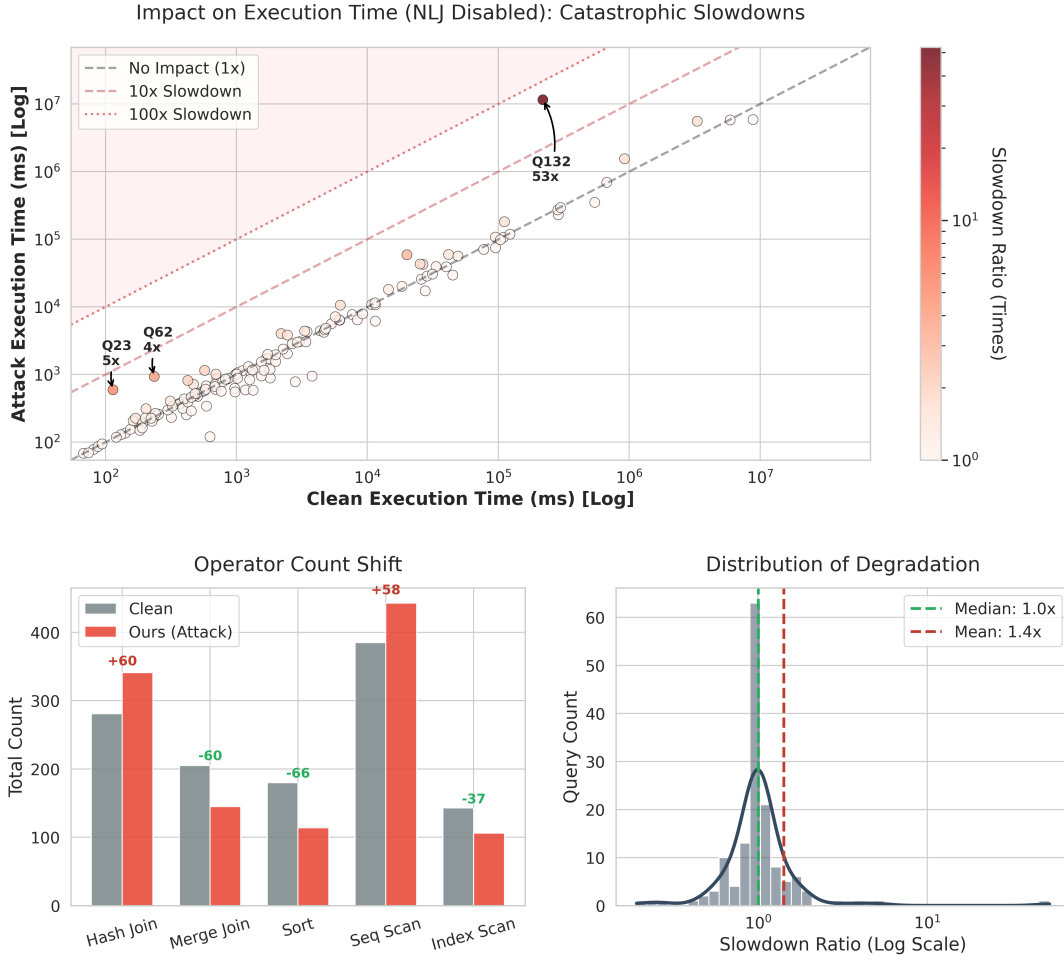


Figure 11: The “Merge Join Trap”: Performance degradation audit with Nested Loop Joins disabled. (Top-Left) The DSA-perturbed estimator (Ours) forces 53× regressions even without NLJs. (Bottom-Left) The failure mechanism shifts from NLJ to expensive Merge Joins and Sorts. (Bottom-Right) Distribution of degradation.

Crucially, to isolate the impact of cardinality estimation quality on query performance, we disabled parallel query execution (setting `max_parallel_workers` and related parameters to 0) and the Genetic Query Optimizer (`geqo = off`). This ensures that the reported end-to-end latencies reflect the efficiency of the physical plans generated by the optimizer, rather than variances introduced by thread scheduling or nondeterministic planning heuristics.

G UTILITY ANALYSIS: THE POTENTIAL& LIMITS OF OPERATOR RESTRICTION

Our primary evaluation (§ 5.2) revealed that many of the catastrophic performance regressions identified by DSA were associated with the optimizer erroneously selecting *Nested Loop Joins* (NLJ) for large intermediate results. This observation suggests a potential heuristic defense: *Can we immunize the system against worst-case data drifts simply by disabling risky operators like NLJ?*

To evaluate the efficacy of this “Safety-First” configuration, we re-conducted the DSA audit on the STATS-CEB benchmark with NLJs explicitly disabled (`enable_nestloop = off`).

G.1 Average-Case Mitigation vs. Tail Latency Failure

The results, visualized in Figure 11, present a nuanced finding. On one hand, enforcing this operator restriction successfully mitigates the impact of estimation errors for the median query, preventing the naive $O(M \times N)$ complexity explosions seen in the baseline settings.

However, the DSA reveals that this heuristic defense is insufficient for the **tail distribution**. As shown in Figure 11 (Top-Left), specific data drifts still trigger catastrophic slowdowns, with the worst-case query (Q132) degrading by 52.77×. This demonstrates that while disabling NLJs acts as a “guardrail” for average cases, it cannot compensate for the fundamental blindness of the estimator regarding specific data correlations.

Table 3: PostgreSQL 13.1 Configuration Parameters (Non-default)

Parameter Name	Value	Unit	Description
<i>Resource Usage & Memory</i>			
shared_buffers	524288	8kB	Shared memory buffers (approx. 4GB)
work_mem	4096	kB	(Implicit default in CEB environment)
wal_buffers	2048	8kB	Disk-page buffers in shared memory for WAL
max_stack_depth	2048	kB	Maximum stack depth
dynamic_shared_memory_type	posix	-	Dynamic shared memory implementation
<i>Planner & Optimizer (Crucial for Reproducibility)</i>			
geqo	off	-	Disable Genetic Query Optimization
max_parallel_workers	0	-	Disable Parallel Execution
max_parallel_workers_per_gather	0	-	Disable parallel processes per node
default_text_search_config	pg_catalog.english	-	Default text search configuration
<i>Write Ahead Log (WAL) & Checkpoints</i>			
max_wal_size	51200	MB	WAL size triggering a checkpoint
min_wal_size	80	MB	Minimum WAL size
wal_segment_size	16777216	B	Size of WAL segments
archive_command	(disabled)	-	Command to archive WAL files
<i>Connections & Networking</i>			
max_connections	100	-	Max concurrent connections
listen_addresses	*	-	Listen on all IP addresses
port	5432	-	Listening port
tcp_keepalives_count	9	-	Max TCP keepalive retransmits
tcp_keepalives_idle	7200	s	Time between TCP keepalives
tcp_keepalives_interval	75	s	Interval between keepalive retransmits
unix_socket_permissions	0777	-	Access permissions for Unix socket
<i>Locale, Formatting & Paths</i>			
DateStyle	ISO, MDY	-	Date and time display format
TimeZone	Etc/UTC	-	Time zone for timestamps
lc_collate	C	-	Collation order locale
lc_ctype	C	-	Character classification locale
server_encoding	SQL_ASCII	-	Database character set encoding
data_directory	/var/lib/pgsql/13.1/data	-	Server data directory
config_file	.../postgresql.conf	-	Main configuration file path

G.2 Mechanism of Defense Failure: The "Merge Join Trap"

By analyzing the execution plans generated under this restrictive policy, we identified a phenomenon we term the **"Merge Join Trap."** When the estimator drastically underestimates cardinality (due to the DSA-targeted drift), and the "easy" path of NLJ is blocked, the optimizer seeks the *next cheapest* alternative based on its flawed cost model.

Typically, it erroneously predicts that the intermediate results are small enough to be sorted and merged efficiently. In reality, the massive data volume triggers disk-spilling Sort operations and high-overhead Merge Joins. This confirms that the root cause of fragility is not the choice of physical operator, but the *data-driven belief system* of the estimator itself.

G.3 Case Study: STATS-Q132

Listing 1 illustrates this failure mode. Even with the "Safety-First" configuration (Plan B), the perturbed estimator misleads the optimizer into a path that is theoretically safe but practically disastrous.

```

1  -- [A] OPTIMAL PLAN (Clean Data, No NLJ)
2  -- Runtime: 219s | Strategy: Hash Join
3  --> Aggregate
4  ...
5  --> Hash Join (p.OwnerUserId = u.Id)
6  -- Optimizer correctly sees high selectivity
7  --> Seq Scan on posts p (Filter: AnsCount<=5)
8  --> Hash (users u)
9
10 -- [B] FAILED DEFENSE PLAN (DSA Perturbed, No NLJ)
11 -- Runtime: 11,562s | Cost: ~7,093
12 -- Pathology: The "Merge Join Trap"
13 --> Aggregate
14 --> Merge Join (v.UserId = u.Id)
15 --> Merge Join (v.UserId = ph.UserId)
16 -- < Est. Rows: Low -> Choice: Merge >
17 --> Merge Join (p.OwnerUserId = v.UserId)

```

```

18 --> Sort (Key: p.OwnerUserId)
19 --> Materialize
20 -- < REALITY: Intermediate result explodes >
21 -- < Actual: 3.8 BILLION rows >
22 --> Index Scan on postHistory ph

```

Listing 1: Comparison of Plans under "Safety-First" Configuration (No NLJ). Even without NLJs, the estimator forces a fallback to a pathological Merge Join.

Takeaways: Strategic Insight: Beyond Heuristic Defenses: This experiment highlights the unique value of **Data-centric Sensitivity Analysis (DSA)**. It proves that reactive heuristics—such as disabling risky operators—are essentially "band-aids" that mask symptoms but do not cure the disease. DSA enables us to move beyond these superficial fixes by exposing the **theoretical robustness limits** of the estimator. It suggests that next-generation robust optimizers must not only restrict operators but also incorporate **uncertainty-aware planning** that can detect and mitigate the specific data-level blind spots revealed by our audit.

H UTILITY ANALYSIS: FURTHER EXPLORATIONS

Following the utility analysis in Section 5.5, this section provides additional explorations into how DSA identifies system vulnerabilities and evaluates potential mechanisms for enhancing the robustness of learned cardinality estimators.

H.1 Hardening Estimators via Controlled Noise Injection

As demonstrated by our DSA auditor, learned models often exhibit significant performance regressions when encountering specific "heavy-hitter" data distributions. To mitigate these worst-case failures, we investigate a robustness-enhancing strategy inspired by uncertainty-aware mechanisms in differential privacy [28, 29, 65]. By systematically introducing controlled perturbations to estimator outputs, we can mask the fine-grained data characteristics that trigger model instability, effectively creating a "buffer zone" against distribution shifts.

Formally, we modify the estimator output to generate a more robust prediction:

$$\text{Est}_{\text{Robust}} = \text{Est} + |\alpha \cdot \eta| \quad (10)$$

where σ denotes the standard deviation of estimator outputs, α is a scaling factor for robustness intensity, and $\eta \sim \mathcal{N}(0, \sigma^2)$ represents Gaussian noise. This approach establishes an $\alpha\sigma$ -radius protection boundary, shielding the downstream optimizer from highly sensitive (and potentially erroneous) cardinality estimates.

Figure 12 illustrates the impact of this hardening strategy across different estimators and datasets. Our diagnostic analysis reveals three key insights into system robustness:

- **Robustness Gains:** Strategic noise injection (reducing Q-Mean by up to three orders of magnitude) successfully mitigates the catastrophic failures identified by our DSA auditor, stabilizing the estimator's performance under stress.
- **Configuration Sensitivity:** The optimal robustness parameter α varies significantly across datasets and model

architectures. This highlights that "blind" hardening is insufficient; precise auditing via DSA is required to determine the appropriate intensity.

- **The Stability-Precision Tradeoff:** Excessive perturbation ($\alpha > 1.0$) leads to model "collapse," where the error introduced by the hardening mechanism itself exceeds the risks posed by data drift. This underscores the importance of the DSA framework in finding the "sweet spot" between nominal precision and worst-case stability.

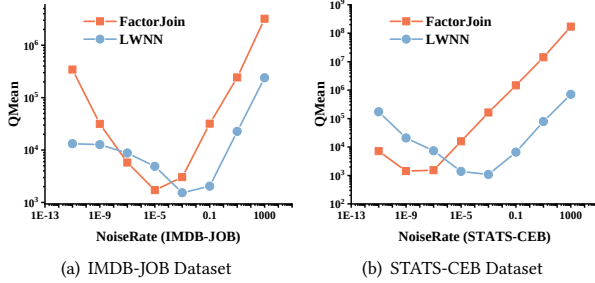


Figure 12: Evaluating Robustness Enhancement via Controlled Noise Injection.

Strategic Insights: Our investigation confirms that building robust database systems requires a rigorous understanding of an estimator’s vulnerability profile. The DSA framework provides the theoretical and empirical foundation to: (1) Characterize the "robustness surface" of learned components; (2) Calibrate noise-response curves for automated system tuning; and (3) Establish practical bounds for the tradeoff between estimation accuracy and execution-plan stability.

I DETAILED SETTINGS FOR ROBUST MODEL MAINTENANCE AUDITING

In this section, we provide the technical specifics and logical motivation for the proactive maintenance experiment discussed in Section 5.5.

I.1 Heavy-Hitter Update Scenario Modeling

To rigorously test the "Early Warning" capability of DSA, we simulate a *worst-case yet realistic* dynamic environment. Traditional evaluations often assume uniform or random data updates; however, real-world database activity is frequently skewed toward specific "Heavy Hitter" tuples—data points that are both frequently updated and highly relevant to join-intensive queries (e.g., trending posts in the STATS-CEB dataset).

We perform 20 consecutive rounds of updates on the posts table. In each round, we modify 1% of the total tuples. Crucially, these modifications are not random but target the top-1% of tuples with the highest joint weights, creating a "correlated drift" that is particularly lethal to outdated learned estimators. This setup represents a scenario where a small but significant shift in key data distributions can lead to catastrophic optimizer failures if not detected early.

I.2 Maintenance Strategy Specifications

We compare the *DSA-Guided Sentinel* against two traditional paradigms:

- **Reactive (Post-hoc):** This strategy mimics a system that lacks proactive monitoring. It monitors the *observed* estimation error during runtime. An update is triggered only after the average Q-error of the executed workload exceeds a threshold of 50. As shown in our results, this strategy inevitably suffers from a "detection lag," as the model is already compromised by the time the error is recorded.
- **Periodic (High Budget):** This is a conservative strategy that ignores data characteristics and retraining the model every 2% of total data updates ($U = 10$ total retrains). While stable, it incurs prohibitive overhead by performing redundant training even when the distribution shift is benign.
- **DSA-Guided Sentinel (Ours):** Leveraging the risk continuity insight (see Appendix L.1), this strategy operates as a precision tripwire. Instead of monitoring all data, we pre-identify a *Critical Set* comprising the top 20% of tuples with the highest joint weights (as approximated by DSA). During the online phase, we track updates exclusively targeting this high-sensitivity subset. A retraining process is triggered only when the cumulative drift impacts 10% of this Critical Set. This approach filters out statistical noise, ensuring maintenance resources are allocated only when structural drifts threaten to trigger the "Merge Join Trap."

I.3 Efficiency Analysis

By focusing on the structural sensitivity of the data distribution rather than raw update volume or historical error, the DSA-Guided Sentinel identifies the *critical mass* of drift. As illustrated in Figure 7, DSA maintains a near-optimal error profile with only 4 retraining operations ($U = 4$), effectively suppressing the catastrophic error spikes seen in the Reactive approach while achieving a 60% reduction in computational budget compared to the Periodic method.

J EXTENDED SENSITIVITY ANALYSIS ON IMDB

To further validate the robustness of our budget calibration across diverse data distributions, we extend our evaluation to the IMDB dataset (using JOB-Light queries). IMDB is widely recognized for its challenging join correlations and non-uniform distributions, providing a rigorous testbed for auditing methods. These results, summarized in Table 4, corroborate the observations from Section 5.2 with even more pronounced performance stratification.

First, at extremely low budgets (e.g., 31,603 rows, representing only **0.05%** of the total database), traditional drift patterns fail to compromise the estimator. The *Sequential* method remains almost entirely inert (Median Q-Error ≈ 1.0), while the *Heuristic* method only induces a marginal median error of 1.90. In contrast, DSA identifies a worst-case perturbation that drives the 95th percentile error to 3.22×10^5 , demonstrating that structural fragility can be exploited even when a negligible fraction of the database is modified.

Second, the data justifies the 20% local budget (0.82% of the total DB) as a reasonable "activation threshold" for baseline methods. At this level, the *Heuristic* approach begins to show non-trivial

degradation (Median Q-Error of 18.6 and Max of 3.24×10^7). However, under the same budget, DSA renders the estimator essentially dysfunctional, with a median error of 1.50×10^5 and a maximum error reaching 9.54×10^9 . This comparison reinforces that while baselines can eventually induce error given enough budget, they lack the precision of DSA in finding the estimator’s most vulnerable regions.

Table 4: Sensitivity Analysis and Budget Calibration on IMDB.

Drift Constraint / Budget			Audit Method	Sensitivity Metric (Q-Error)			
Rows	Table %	Total DB %		50%	90%	95%	Max
31,603	1.25%	0.05%	Sequential	1.00	1.01	1.01	1.01
			Heuristic	1.90	14.4	48.1	715
			DSA (Ours)	9.14	1.55×10^5	3.22×10^5	3.24×10^7
63,206	2.50%	0.10%	Sequential	1.06	1.07	1.08	1.09
			Heuristic	2.39	42.6	244	1.14×10^6
			DSA (Ours)	16.4	2.72×10^5	8.83×10^5	3.24×10^7
126,412	5.00%	0.20%	Sequential	1.14	1.17	1.18	1.25
			Heuristic	3.70	110	509	1.50×10^6
			DSA (Ours)	33.2	9.03×10^6	4.28×10^7	1.35×10^9
252,824	10.0%	0.41%	Sequential	1.44	1.59	1.83	3.20
			Heuristic	6.81	458	1.82×10^3	1.50×10^6
			DSA (Ours)	77.9	2.73×10^7	8.30×10^7	1.35×10^9
505,648	20.0%	0.82%	Sequential	2.15	2.44	2.77	780
			Heuristic	18.6	1.82×10^3	1.24×10^4	3.24×10^7
			DSA (Ours)	1.50×10^5	1.23×10^8	5.04×10^8	9.54×10^9

Takeaways: On the IMDB dataset, the gap between targeted and heuristic drifts is even more pronounced than on STATS. DSA reveals that cardinality estimators can be driven to catastrophic failure (95th percentile error $> 10^8$) at budgets as low as 0.82% of the total database, confirming that structural vulnerabilities are a general property of learned estimators across complex schemas.

K WHY THE ORACLE OBJECTIVE IS INFORMATIVE FOR LEARNED CARDINALITY ESTIMATORS

What this appendix is for. Our auditing objective (DSA) searches for a perturbed snapshot D' that makes the oracle cardinality on D' disagree as much as possible with the truth on the original snapshot D for a target workload W . The main text uses this oracle-based objective as a proxy for failures of learned cardinality estimators trained on D' . This appendix provides a compact but formal justification:

If a learned estimator fits the perturbed snapshot D' within a multiplicative factor α on the workload, then its drift-induced error on D is within an α -factor of the oracle discrepancy induced by the same perturbation.

Concretely, we (i) establish basic composition rules of Q-error, (ii) derive an oracle $\leftrightarrow$ learned error transfer theorem, and (iii) provide a general theoretical guarantee showing that any model minimizing log-space loss inherently optimizes α , followed by a case study linking parameter convergence to risk for linear representations.

K.1 Q-error as a log-metric

Why smoothing is needed. Cardinality queries may produce zero counts. To avoid degeneracies and keep all expressions well-defined,

we use a standard smoothed Q-error with a small $\tau > 0$ (e.g., $\tau = 1$), which is common in .

Definition K.1 (Smoothed Q-error). For $\tau > 0$ and any $a, c \geq 0$, define

$$\text{QErr}_\tau(a, c) = \max \left\{ \frac{a + \tau}{c + \tau}, \frac{c + \tau}{a + \tau} \right\}.$$

When $a, c > 0$ and $\tau = 0$, this reduces to the standard $\text{QErr}(a, c)$.

Key idea. Q-error becomes particularly clean when viewed in log space: it is simply an exponentiated absolute difference. This lets us use the triangle inequality to “compose” errors, which will be the engine behind the oracle-to-learned transfer bound.

Definition K.2 (Log-distance). For any $x, y > 0$, define

$$d(x, y) = |\log x - \log y|.$$

LEMMA K.3 (Q-ERROR EQUALS EXPONENTIATED LOG-DISTANCE).

For any $a, c \geq 0$ and $\tau > 0$,

$$\text{QErr}_\tau(a, c) = \exp(d(a + \tau, c + \tau)).$$

PROOF. Let $x = a + \tau > 0$ and $y = c + \tau > 0$. Then

$$d(x, y) = |\log x - \log y| = \log \left(\max \left\{ \frac{x}{y}, \frac{y}{x} \right\} \right).$$

Exponentiating both sides yields

$$\exp(d(x, y)) = \max \left\{ \frac{x}{y}, \frac{y}{x} \right\} = \text{QErr}_\tau(a, c).$$

□

Takeaways: Lemma K.3 turns Q-error into a metric-like quantity in log space. The next two lemmas convert this into practical algebraic rules: errors multiply (triangle inequality) and can also be lower-bounded (reverse triangle inequality).

K.2 Two composition rules for Q-error

LEMMA K.4 (MULTIPLICATIVE TRIANGLE INEQUALITY). For any $a, b, c \geq 0$ and $\tau > 0$,

$$\text{QErr}_\tau(a, c) \leq \text{QErr}_\tau(a, b) \cdot \text{QErr}_\tau(b, c).$$

(Intuition). Given Q-errors a, b, c , If a is close to b (small Q-error) and b is close to c , then a cannot be far from c —and the resulting distortion is multiplicative.

PROOF. Let $x = a + \tau, y = b + \tau, z = c + \tau$, all positive. By Lemma K.3,

$$\log \text{QErr}_\tau(a, c) = d(x, z), \quad \log \text{QErr}_\tau(a, b) = d(x, y), \quad \log \text{QErr}_\tau(b, c) = d(y, z).$$

Since $d(\cdot, \cdot)$ is the absolute difference after applying $\log(\cdot)$, it is a metric on $(0, \infty)$ and satisfies the (additive) triangle inequality:

$$d(x, z) \leq d(x, y) + d(y, z).$$

Exponentiating gives

$$\text{QErr}_\tau(a, c) = \exp(d(x, z)) \leq \exp(d(x, y)) \exp(d(y, z)) = \text{QErr}_\tau(a, b) \cdot \text{QErr}_\tau(b, c).$$

□

LEMMA K.5 (REVERSE MULTIPLICATIVE BOUND). For any $a, b, c \geq 0$ and $\tau > 0$,

$$\text{QErr}_\tau(a, c) \geq \frac{\text{QErr}_\tau(b, c)}{\text{QErr}_\tau(a, b)}.$$

(Intuition). If b is far from c , then a must also be far from c unless a is already far from b . “This lemma will give us a clean lower bound relating learned error to oracle discrepancy.”

PROOF. Let $x = a + \tau$, $y = b + \tau$, $z = c + \tau$. By Lemma K.3, $\log \text{QErr}_\tau(a, c) = d(x, z)$ etc. The reverse triangle inequality for a metric gives:

$$d(x, z) \geq |d(y, z) - d(x, y)| \geq d(y, z) - d(x, y).$$

Exponentiating yields

$$\text{QErr}_\tau(a, c) = \exp(d(x, z)) \geq \exp(d(y, z) - d(x, y)) = \frac{\exp(d(y, z))}{\exp(d(x, y))} = \frac{\text{QErr}_\tau(b, c)}{\text{QErr}_\tau(a, b)}.$$

□

Takeaways: Lemmas K.4–K.5 provides “math tools” we need: they formalize how multiplicative errors propagate across two steps. We now apply them to connect (i) the oracle discrepancy caused by drifting from D to D' and (ii) the learned estimator error after being trained on D' .

K.3 Oracle $\rightarrow$ learned transfer bound

Setting. Let D be the original snapshot and D' be a perturbed snapshot. For a query q , let $C_X(q)$ denote the true cardinality on snapshot X . Let $\widehat{C}_{D'}(q)$ be a learned estimator trained on D' . We focus on a workload W .

Formalizing the missing link. The main text utilizes an oracle-based objective of the form $\text{QErr}_\tau(C_{D'}(q), C_D(q))$. To extend this to average learned CEs, we ground our analysis in a consistent empirical property observed in recent cardinality estimation literature: the Q-error distribution is typically heavily skewed toward 1 (often covering $> 90\%$ of queries), while the maximum Q-errors are generally confined to a manageable magnitude (e.g., typically within double digits) [38, 56, 66, 72, 77, 88]. Motivated by this, we introduce α to explicitly model this worst-case boundary on the target database D' .

Definition K.6 (performance factor of a learned CE on D'). For $\tau > 0$, let α denote the maximum Q-error over the workload W :

$$\alpha = \max_{q \in W} \text{QErr}_\tau(\widehat{C}_{D'}(q), C_{D'}(q)).$$

Equivalently, $\forall q \in W$, $\text{QErr}_\tau(\widehat{C}_{D'}(q), C_{D'}(q)) \leq \alpha$.

The quantity α effectively captures the learned model’s fidelity to D' . High-capacity learned CEs are explicitly optimized to minimize α on their training distribution. The subsequent theorem demonstrates that once α is bounded, the oracle objective provides direct control over the learned CE’s failure probability under data drift.

THEOREM K.7 (ORACLE-TO-LEARNED ERROR TRANSFER (POINTWISE)). Fix any $\tau > 0$ and any query $q \in W$. If $\text{QErr}_\tau(\widehat{C}_{D'}(q), C_{D'}(q)) \leq \alpha$, then

$$\frac{1}{\alpha} \text{QErr}_\tau(C_{D'}(q), C_D(q)) \leq \text{QErr}_\tau(\widehat{C}_{D'}(q), C_D(q)) \leq \alpha \text{QErr}_\tau(C_{D'}(q), C_D(q)).$$

PROOF. Let $a = \widehat{C}_{D'}(q)$, $b = C_{D'}(q)$, and $c = C_D(q)$.

Upper bound. By Lemma K.4,

$$\text{QErr}_\tau(a, c) \leq \text{QErr}_\tau(a, b) \cdot \text{QErr}_\tau(b, c) \leq \alpha \cdot \text{QErr}_\tau(b, c),$$

where the last step uses $\text{QErr}_\tau(a, b) \leq \alpha$.

Lower bound. By Lemma K.5,

$$\text{QErr}_\tau(a, c) \geq \frac{\text{QErr}_\tau(b, c)}{\text{QErr}_\tau(a, b)} \geq \frac{\text{QErr}_\tau(b, c)}{\alpha},$$

again using $\text{QErr}_\tau(a, b) \leq \alpha$. Combining both inequalities completes the proof. □

What this resolves. Theorem K.7 states that the learned estimator’s drift-induced error on D is sandwiched by the oracle discrepancy up to a factor α . In other words, when a learned estimator fits D' well (small α), maximizing the oracle objective is effectively maximizing the learned estimator’s worst-case risk as well.

COROLLARY K.1 (WORKLOAD AGGREGATE ERROR TRANSFER). Under Definition K.6, for any $\tau > 0$, the aggregate Q-errors satisfy:

$$\begin{aligned} & \frac{1}{\alpha} \sum_{q \in W} \text{QErr}_\tau(C_{D'}(q), C_D(q)) \\ & \leq \sum_{q \in W} \text{QErr}_\tau(\widehat{C}_{D'}(q), C_D(q)) \\ & \leq \alpha \sum_{q \in W} \text{QErr}_\tau(C_{D'}(q), C_D(q)). \end{aligned} \quad (11)$$

PROOF. This follows by summing the pointwise inequalities established in Theorem K.7 over all $q \in W$. Consider the upper bound:

$$\sum_{q \in W} \text{QErr}_\tau(\widehat{C}_{D'}(q), C_D(q)) \leq \sum_{q \in W} [\alpha \cdot \text{QErr}_\tau(C_{D'}(q), C_D(q))].$$

Since α is a constant with respect to the query q , it factors out of the summation, yielding the right-hand side. The lower bound follows symmetrically. □

Takeaways: Corollary K.1 is the exact statement we need for DSA: an oracle-based audit objective is meaningful for learned estimators as soon as the learned estimator achieves bounded fit α on the perturbed snapshot.

K.4 Theoretical Guarantee: From Log-Space Loss to α

Overview. In this section, we provide the theoretical justification for the learned estimator’s performance factor α . We first establish a **model-agnostic theorem** showing that any estimator trained to minimize error in log-space inherently minimizes the Q-error bound α . We then provide a **concrete case study** on linear representations (common in deep learning last-layers) to show how parameter convergence ensures this bound is met.

1. General Principle. Most modern learned cardinality estimators (e.g., MSCN, NeuroCard) operate by predicting log-cardinalities. The following theorem proves that this common practice is mathematically equivalent to optimizing the worst-case Q-error α .

THEOREM K.8 (LOG-SPACE ERROR CONTROLS Q-ERROR). Let $\widehat{y}_{D'}(q)$ be any predictor for the smoothed log-cardinality $y_{D'}(q) = \log(C_{D'}(q) + \tau)$. If the estimator achieves a bounded error in log-space, i.e.,

$$\forall q \in W, \quad |\widehat{y}_{D'}(q) - y_{D'}(q)| \leq \eta,$$

then the Q -error is bounded by e^η . Consequently, the workload fit factor satisfies:

$$\alpha \leq e^\eta.$$

PROOF. By definition, $\widehat{C}_{D'}(q) = \exp(\widehat{y}_{D'}(q)) - \tau$. Using the property that $QErr_\tau(A, B) = \exp(|\log(A + \tau) - \log(B + \tau)|)$, we have:

$$QErr_\tau(\widehat{C}_{D'}(q), C_{D'}(q)) = \exp(|\widehat{y}_{D'}(q) - y_{D'}(q)|).$$

Substituting the condition $|\widehat{y}_{D'}(q) - y_{D'}(q)| \leq \eta$, we get $QErr_\tau \leq e^\eta$. Since this holds for all $q \in W$, the maximum α is also bounded by e^η . $\square$

Significance. Theorem K.8 is universal. It implies that for any model architecture (regression trees, deep neural networks, etc.), minimizing the L_∞ or MSE loss in log-space is a theoretically valid strategy to control α .

2. Case Study: Linear Representations. While Theorem K.8 gives a condition on the output η , we now analyze how this condition is achieved in practice via parameter training. We focus on models where the prediction is linear over a feature representation $\phi(q)$ (which captures both linear regression and the final layer of deep neural networks) [48, 56, 60].

PROPOSITION K.9 (PARAMETER CONVERGENCE IMPLIES BOUNDED α). Consider a model $\widehat{y}_{D'}(q) = w^\top \phi(q)$ where features are bounded ($\|\phi(q)\|_2 \leq B$). Assume the target concept is realizable by some w^* . If the learned parameters w are close to optimal such that $\|w - w^*\|_2 \leq \delta$, then the condition of Theorem K.8 is satisfied with $\eta = \delta B$. Thus:

$$\alpha \leq \exp(\delta B).$$

PROOF. The log-space error is $|\widehat{y}_{D'}(q) - y_{D'}(q)| = |(w - w^*)^\top \phi(q)|$. By Cauchy-Schwarz inequality and linearity, this error is bounded by $\|w - w^*\|_2 \|\phi(q)\|_2 \leq \delta B$. Applying Theorem K.8 with $\eta = \delta B$ completes the proof. $\square$

Takeaways: Theorem K.8 confirms that any model successfully trained in log-space is a valid proxy for the oracle (small α). Proposition K.9 demonstrates that for standard architectures, this success is directly tied to parameter convergence (δ).

L EXTENDED DISCUSSION: PRACTICAL NECESSITY AND REAL-WORLD MAPPING OF DSA

In this section, we further elucidate the practical necessity of our proposed worst-case analysis framework. Our investigation reveals three critical insights regarding the deployment of learned estimators in production: (1) **Utility of Partial Overlap:** The worst-case subsets found by DSA are highly actionable; any data update that partially overlaps with these subsets is sufficient to trigger degradation, establishing the theoretical basis for our Sentinel defense. (2) **High Realistic Probability:** The distributions identified by DSA possess high plausibility in business contexts, mapping to structural business risks. (3) **Engineering Assurance:** Unlike systems that ignore worst-case scenarios, DSA provides the necessary early warnings to satisfy strict Service Level Agreements (SLAs).

L.1 Risk Continuity: The Foundation of Sentinel Defense

We find that the worst-case subsets identified by DSA are robust indicators of system fragility. Due to the greedy nature of our approximation strategy (Algorithm 1), the perturbation set is constructed sequentially based on marginal impact magnitude. Consequently, the risk is **continuous and monotonic**: a real-world data drift does not need to perfectly replicate the worst-case scenario to cause damage. **Key Insight:** As long as a potential data update overlaps—even partially—with the high-risk tuples identified by DSA, the estimator’s accuracy suffers significant degradation.

Practical Implication. This property validates that DSA functions as a powerful ally for system monitoring. Since the “Top- K ” perturbed tuples contribute disproportionately to the error, they serve as the most effective watch-list. This theoretical finding underpins our *Sentinel* experiment (Section 5.5): by monitoring updates to only these critical tuples, the system can detect and preempt performance regression efficiently. Thus, DSA acts not merely as a theoretical auditor, but as a **continuous quantification metric** that guides proactive maintenance.

L.2 From Mathematical Extremes to Business Reality

From a database business perspective, the data distributions identified by DSA possess high realistic plausibility. By analyzing the target tuples locked by DSA, we observe a distinct pattern: these tuples are typically characterized by high joint weights and frequent participation in complex joins. In real-world scenarios, these are not random outliers, but map directly to **High-Value Data**:

- **E-commerce:** “Best-selling items” or “Flash-sale products” (accessed by thousands of customers simultaneously).
- **Social Networks:** “Influencers” or “Viral Posts” (generating massive fan-out in join graphs).
- **Finance:** “Core Institutional Accounts” (involved in high-frequency transactions).

Key Insight: The data points most destructive to the model (as identified by DSA) are often exactly those that are most active and dynamic in real business logic. Therefore, the scenario simulated by DSA is a precise abstraction of **Structured Business Risks**, rather than a rare coincidence.

L.3 Engineering Principles: The “Bottom-Line” Guarantee

Finally, we address the necessity of DSA through the lens of System Reliability Engineering (SRE). Critical infrastructure requires a “Zero-Tolerance” principle for blind spots.

The value of DSA lies in providing a **Theoretical Performance Floor**. If a learned system rejects worst-case analysis, it exposes itself to “Black Swan” latency spikes (e.g., the Merge Join Trap) without warning, making it impossible to guarantee Service Level Agreements (SLAs). Conversely, DSA serves as the necessary *Stress Test* to define safety margins. By quantifying the theoretical lower bound, DSA acts as an early-warning system, certifying whether the estimator can sustain strict SLAs (e.g., P99 latency) under stress

conditions. This transforms reliability from a gamble into an engineered guarantee.

Summary: (1) **Actionable Utility:** Partial drifts on DSA-identified tuples trigger significant degradation, validating the efficiency of “Sentinel” monitoring. (2) **Business Relevance:** The mathematical worst-case highly correlates with real-world “Heavy Hitters,” making the threat structurally plausible. (3) **SLA Support:** DSA defines the theoretical performance floor, providing the essential pre-warning required to meet strict SLAs.